

Exploratory Research Project: Few-Shot Generative Modelling Architectures Analysis

Report submitted in fulfillment of the requirements
for the Exploratory Project of
Second Year B.Tech.

by

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Dedicated to

Our parents, our teachers, and the relentless pursuit of knowledge.

Declaration

We hereby certify that:

1. The work contained in this report is original and has been executed by us under the general supervision of our designated supervisor.
2. The research and writing presented herein have not been submitted, either in part or in full, for any other project, degree, or diploma at this or any other institution.
3. Whenever materials (data, theoretical analysis, empirical results) from external sources have been utilized, appropriate credit has been attributed by citing them within the text and providing full details in the reference section.
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Certificate

This is to certify that the exploratory research project entitled “**Few-Shot Generative Modelling Architectures Analysis**” being submitted by **Vishweshvar RaviShankar K (Roll No: 24074035)**, **Shivang Chaudhary(Roll No: 24075082)**, and **Gagan Siddartha (Roll No: 24075018)**, carried out in the Department of Computer Science and Engineering, Indian Institute of Technology (BHU) Varanasi, is a bona fide record of work completed under our direct supervision and guidance.

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Place: IIT (BHU) Varanasi

Date: April 19, 2026

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Abstract

Generative modelling has achieved unprecedented success in synthesizing high-fidelity data; however, these models inherently demand vast amounts of training data, rendering them impractical for domains suffering from severe data scarcity. This report conducts a comprehensive exploratory analysis of Generative Modelling under Data Constraint (GM-DC). Specifically, we rigorously evaluate a diverse spectrum of state-of-the-art architectures operating under extreme limited-data regimes from scratch (uGM-1) and few-shot transfer learning regimes (uGM-2).

The evaluated frameworks encompass structural freezing baselines (FreezeD), adaptive augmentation and noise injection (StyleGAN2-ADA, DANI), contrastive instance discrimination (FakeCLR, InsGen, DCL), cross-domain consistency regularization (CDC, NICE), and targeted knowledge truncation (RICK). By tracking structural fidelity against internal variance utilizing metrics such as Fréchet Inception Distance (FID), Precision, Recall, Density, Coverage, and Pairwise LPIPS, this research exposes the underlying mechanics of mode collapse and overfitting in constrained regimes. The findings indicate that while mechanisms like noise modulation, contrastive learning, and filter truncation successfully stabilize training and preserve realism, maintaining uniform target domain diversity (Recall) remains a critical, overarching bottleneck across all few-shot generative adaptations.

Contents

Declaration	ii
Certificate	iii
Acknowledgments	iv
Abstract	v
List of Figures	viii
List of Symbols	x
1 Introduction	1
1.1 Our Contributions	1
1.2 Background	2
1.3 Challenges in GM-DC	2
1.4 Evaluation Metrics	3
2 Architecture 1: StyleGAN2-ADA	4
2.1 Overview and Methodology	4
2.2 Performance Analysis	5
2.3 Qualitative Analysis	8
2.4 Conclusion and Final Metrics	8
3 Architecture 2: DANI (Diffusion-GAN)	9
3.1 Overview and Methodology	9
3.2 Performance Analysis	10
3.3 Qualitative Analysis	12
3.4 Conclusion and Final Metrics	13
4 Architecture 3: FakeCLR	14
4.1 Overview and Methodology	14
4.2 Performance Analysis	15
4.3 Qualitative Analysis	17
4.4 Conclusion and Final Metrics	18
5 Architecture 4: InsGen	19
5.1 Overview and Methodology	19
5.2 Performance Analysis	20
5.3 Qualitative Analysis	23
5.4 Conclusion and Final Metrics	23
6 Architecture 5: NICE	24

6.1	Overview and Methodology	24
6.2	Performance Analysis	25
6.3	Qualitative Analysis	28
6.4	Conclusion and Final Metrics	29
7	Architecture 6: RICK	30
7.1	Overview and Methodology	30
7.2	Performance Analysis	31
7.3	Qualitative Analysis	34
7.4	Conclusion and Final Metrics	35
8	Architecture 7: FreezeD	36
8.1	Overview and Methodology	36
8.2	Performance Analysis	37
8.3	Qualitative Analysis	38
8.4	Conclusion and Final Metrics	39
9	Architecture 8: CDC	40
9.1	Overview and Methodology	40
9.2	Performance Analysis	42
9.3	Qualitative Analysis	42
9.4	Conclusion and Final Metrics	43
10	Architecture 9: DCL	44
10.1	Overview and Methodology	44
10.2	Performance Analysis	45
10.3	Qualitative Analysis	46
10.4	Conclusion and Final Metrics	47
11	Conclusion and Architectural Insights	48
11.1	Summary of the Exploratory Research	48
11.2	Comprehensive Quantitative Comparison	48
11.3	Key Learnings and Architectural Insights	49
11.4	Final Concluding Remarks	50
	References	51

List of Figures

1	The StyleGAN2-ADA pipeline. ADA stochastically applies augmentations (blue blocks) to both real and generated inputs with probability p prior to discriminator evaluation.	4
2	Distribution Distances tracking domain alignment for StyleGAN2-ADA. .	5
3	Correlated tracking of Precision, Recall, and Inception Score.	6
4	Perceptual variance mapping internal structural diversity.	6
5	Final output samples generated by the StyleGAN2-ADA architecture. . .	8
6	The DANI framework pipeline. An Adaptive Forward Diffusion Process injects Gaussian noise into both real (x) and generated ($G(z)$) images simultaneously. The noise probability p is dynamically controlled by the discriminator's overfitting degree (η).	9
7	Distribution Distances tracking domain alignment for DANI over 12 kmg. .	10
8	Correlated tracking of Precision, Recall, and Inception Score.	11
9	Perceptual variance metrics mapping internal structural diversity.	11
10	Final output samples generated by the DANI architecture at 12 kmg. . .	12
11	The FakeCLR framework. Noise-related latent augmentation (blue) constructs positive pairs (v_q, v_q^+), while a diversity-aware queue (red) supplies negative samples ($v_{k_i}^-$). A forgetting factor (green) dynamically scales the gradients based on the training iteration.	14
12	Distribution Distances tracking domain alignment for FakeCLR.	15
13	Correlated tracking of Precision, Recall, and Inception Score.	16
14	Perceptual variance mapping internal structural diversity.	16
15	Final output samples generated by the FakeCLR architecture.	17
16	Conceptual state diagram of InsGen. Training alternates between frozen networks (a and b). The auxiliary instance discrimination task projects features onto a hypersphere, pushing individual sample embeddings ($\phi(v)$) apart to force visual diversity.	19
17	Distribution Distances tracking domain alignment for InsGen.	20
18	Correlated tracking of Precision, Recall, and Inception Score.	21
19	Perceptual variance mapping internal structural diversity.	21
20	Final output samples generated by the InsGen architecture.	23
21	The NICE discriminator pipeline. <i>Left</i> : Adaptive noise modulates latent features; noise variance β is adjusted based on overfitting. <i>Right</i> : Two feature extractors with shared weights but independent noise modulations produce outputs; minimizing their squared difference penalizes gradients to stabilize training. Adapted from [10].	24
22	FID and KID tracking distribution alignment over training.	25
23	Precision and Density plots for NICE.	26
24	LPIPS and MS-SSIM tracing mode collapse and subsequent recovery. . .	26
25	Recall and Coverage exposing the asymmetric nature of generated diversity. .	27

26	NICE-generated samples at successive checkpoints. Early checkpoints exhibit near-uniform collapse, while later checkpoints produce recognizable outputs.	28
27	The RICK pipeline. Filter importance is estimated dynamically from on-the-fly gradient information. Each filter is assigned an operation stored in the memory bank: prune, preserve, or fine-tune.	30
28	Distribution alignment tracking for RICK over training.	31
29	Precision and Density trajectories.	32
30	LPIPS and MS-SSIM tracking perceptual and structural diversity.	32
31	Recall and Coverage tracing diversity decay.	33
32	Radial Power Spectrum (log scale) comparing real and generated images.	33
33	RICK-generated samples at successive training checkpoints. Incompatible FFHQ features are absent throughout.	34
34	The FreezeD framework workflow. Lower convolutional layers of the discriminator are frozen to preserve generic feature extraction, while higher layers remain trainable to adapt to the target distribution.	36
35	FID trajectory comparing standard fine-tuning against the FreezeD architecture.	37
36	Final output samples generated by the FreezeD architecture.	38
37	The CDC framework. A source model (G_s) guides the few-shot adaptation of a target model ($G_{s \rightarrow t}$). Consistency is enforced via a KL-Divergence penalty on relative pairwise distances.	41
38	Final output samples generated by the CDC architecture.	42
39	The DCL framework. Contrastive learning is applied to feature representations, utilizing an InfoNCE objective to actively distinguish between similar (positive) and dissimilar (negative) sample embeddings.	44
40	Final output samples generated by the DCL architecture.	46

List of Symbols

Core Generative Adversarial Networks

P_{data}	True marginal probability distribution of the real target data
P_z	Prior probability distribution of the latent noise space
$G(\cdot)$	Generator network mapping latent vectors to the image space
$D(\cdot)$	Discriminator network evaluating the realism of an image
x, z	A real image sample and a random latent noise vector
$V(D, G)$	The fundamental two-player min-max adversarial value function
μ, Σ	Mean and covariance matrices of extracted feature embeddings (used in FID)

Transfer Learning & Adaptation

G_s, D_s	Pre-trained source generator and discriminator networks
$G_{s \rightarrow t}$	Adapting target generator network
D_{img}, D_{patch}	Image-level (global) and Patch-level (local) dual discriminators
$\mathbf{W}$	Convolutional filter weights within the network
$\mathcal{F}(\mathbf{W})$	Fisher Information metric evaluating convolutional filter importance

Contrastive & Consistency Regularization

$h(\cdot), f(\cdot)$	Feature extractor projection heads mapping to the contrastive space
$\text{sim}(\cdot, \cdot)$	Pairwise cosine similarity function
τ	Temperature scaling hyperparameter for InfoNCE objectives
λ, γ	Scalar weights controlling auxiliary loss contributions
q, k^+, k^-	Query, positive match, and negative match samples in contrastive queuing
D_{KL}	Kullback-Leibler Divergence penalty
$\mathcal{L}_{\text{dist}}$	Distance-preserving cross-domain consistency loss

Noise & Augmentation

p	Dynamic probability of applying stochastic augmentation or diffusion
β	Adaptive noise variance parameter
$\mathbf{Z}$	Random noise modulation tensor drawn from a Bernoulli distribution
η	Discriminator overfitting threshold signal controlling modulation
ϵ	Stochastic noise perturbation injected into the latent vector

1. Introduction

Generative models have emerged as one of the most significant developments in modern machine learning due to their ability to synthesize realistic and diverse data samples. Applications of generative modelling include image synthesis, style transfer, super-resolution, data augmentation, content generation, and simulation tasks. Popular generative frameworks include Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), Autoregressive Models, and Diffusion Models. These models typically require large-scale, high-quality datasets to learn the underlying data distribution effectively.

However, in many real-world scenarios, collecting sufficiently large datasets is expensive, time-consuming, or impractical. Domains such as healthcare imaging, scientific research, remote sensing, historical archives, rare species datasets, and personalized content generation often suffer from severe data scarcity. This challenge has led to increasing research interest in Generative Modelling under Data Constraint (GM-DC).

GM-DC refers to the study and development of generative models capable of producing high-quality outputs when trained with limited, few-shot, or even zero-shot data. Instead of relying on abundant training samples, these methods aim to transfer prior knowledge, regularize learning, stabilize optimization, or exploit pretrained representations to compensate for insufficient data availability.

The survey paper “A Survey on Generative Modelling with Limited Data, Few-Shot, and Zero-Shot” [1] provides a comprehensive categorization of methods designed for such constrained settings. It highlights three major paradigms:

- **Limited Data Generation:** Models trained on small but non-trivial datasets.
- **Few-Shot Generation:** Models adapted using extremely few samples (e.g., 10–100 images).
- **Zero-Shot Generation:** Models generating outputs for unseen domains using prior semantic knowledge.

1.1 Our Contributions

This project goes beyond reproducing reported FID values by presenting a systematic multi-metric evaluation of nine GM-DC architectures.

- **Unified Benchmark with Extended Metrics:** Existing work evaluates GM-DC methods inconsistently, often relying only on FID and occasionally LPIPS. We standardize evaluation across all nine architectures using eight metrics (FID, KID, LPIPS, MS-SSIM, Precision, Recall, Density, Coverage), enabling direct comparison across multiple dimensions, especially Precision–Recall and Density–Coverage, which are rarely reported together.

- **Empirical Validation of the FID Illusion:** Our results show that FID alone can be misleading in constrained settings. Architectures achieve robust FID values but have a very low Recall indicating severe mode collapse. This provides clear experimental evidence that diversity-aware metrics are essential for reliable GM-DC evaluation.

1.2 Background

Generative modelling aims to learn the underlying probability distribution of a dataset so that new samples can be generated from it. For a domain $D = \{X, P_{data}\}$, where X is the sample space and P_{data} is the real data distribution, the objective is to train a model G that learns a distribution P_{model} such that:

$$P_{model} \approx P_{data} \quad (1)$$

Once trained, the generator can produce synthetic samples $x \sim P_{model}$ that statistically resemble the real data.

1.3 Challenges in GM-DC

Training generative models with insufficient data introduces several difficulties that are less severe in large-data regimes.

- **Overfitting:** With very few training images, the generator may memorize the training samples rather than learning a generalized distribution. Generated outputs may closely resemble training images, reducing novelty and diversity.
- **Discriminator Dominance in GANs:** In GAN-based models, the discriminator can easily overfit small datasets and perfectly separate real from fake samples. This prevents the generator from receiving meaningful gradients, causing unstable training.
- **Mode Collapse:** The generator may repeatedly produce limited variations of samples while ignoring other modes of the true data distribution.
- **Poor Diversity:** Even when image quality is visually acceptable, outputs may lack variation in pose, texture, structure, or content.
- **Domain Shift in Transfer Learning:** Few-shot methods often rely on pretrained source models. If the source and target domains differ significantly, incompatible prior knowledge may degrade generation quality.
- **Training Instability:** Small datasets amplify optimization sensitivity, causing oscillation, divergence, or inconsistent convergence.

1.4 Evaluation Metrics

To quantify performance, we employ a multi-dimensional suite of metrics that assess both Fidelity (Quality) and Diversity (Variety).

- **Fréchet Inception Distance (FID):** The primary quality metric in this domain. It computes the Fréchet distance between the distributions of Inception-v3 feature embeddings extracted from real and generated images. A lower FID indicates that the generated distribution is closer to the real one in feature space, reflecting both realism and diversity.

$$\text{FID} = \|\mu_r - \mu_g\|_2^2 + \text{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2}) \quad (2)$$

Where μ and Σ represent the mean and covariance of real (r) and generated (g) image features, and $\text{Tr}(\cdot)$ is the trace of the matrix.

- **Kernel Inception Distance (KID):** A complementary quality metric that uses an unbiased estimator of the maximum mean discrepancy (MMD) between real and generated Inception features. Unlike FID, KID does not assume Gaussian distributions and is more reliable when the number of generated samples is small.
- **LPIPS (Learned Perceptual Image Patch Similarity):** Measures perceptual similarity between pairs of images. Intra-LPIPS—the average perceptual distance among pairs of generated images—is used as a diversity metric. Higher intra-LPIPS values indicate that generated images are more perceptually distinct.
- **MS-SSIM (Multi-Scale Structural Similarity Index):** Assesses structural similarity across multiple resolution scales. Low MS-SSIM indicates high diversity (structurally dissimilar), while high MS-SSIM signals mode collapse.
- **Precision:** Measures the proportion of generated samples that lie within the support region of real data. It indicates how realistic and valid the generated images are.
- **Recall:** Measures how much of the real data distribution is covered by the generated samples, reflecting the diversity and completeness of generation. Low recall suggests mode collapse.
- **Density:** Evaluates how strongly generated samples populate the neighborhoods of real samples, indicating better local realism.
- **Coverage:** Measures the fraction of real samples that have nearby generated samples. It indicates how well the generator represents the overall real dataset.

Importance of Using Multiple Metrics: No single metric fully captures generation performance. Low FID may hide low diversity, high precision may mask memorization, and high diversity alone does not guarantee realism. Therefore, a combination of distributional, perceptual, and manifold-based metrics is necessary for robust comparison.

2. Architecture 1: StyleGAN2-ADA

2.1 Overview and Methodology

The StyleGAN2-ADA architecture is explicitly designed to stabilize generative training in highly constrained, low-data regimes without requiring structural modifications to the core loss formulations. It achieves this through Adaptive Discriminator Augmentation (ADA).

- **Stochastic Augmentation and Leakage Prevention:** ADA subjects both the real training images and the generated samples to an identical pipeline of image-level transformations before evaluation. Let $\mathcal{T}_p$ denote a set of transformations applied non-deterministically with probability p . To prevent "augmentation leakage"—where the generator outputs corrupted artifacts instead of true semantic features—the transformations are designed to be invertible in expectation, forcing the generator to implicitly model the clean underlying data distribution.
- **Dynamic Overfitting-Driven Modulation:** The augmentation strength is not static. The architecture dynamically scales the transformation probability p in real-time based on the discriminator’s degree of overfitting. By continuously monitoring the fraction of training data that the discriminator correctly classifies as real, the system adjusts p to regulate the discriminator’s advantage, stabilizing learning dynamics.

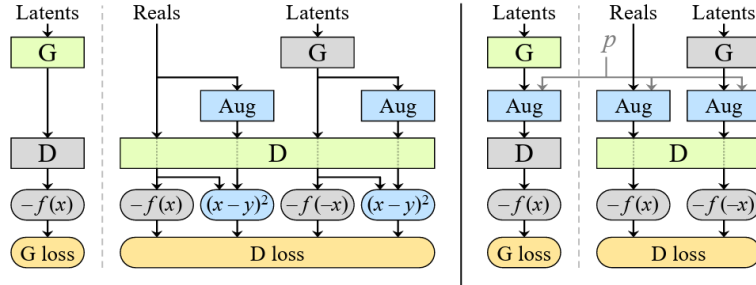


Figure 1: The StyleGAN2-ADA pipeline. ADA stochastically applies augmentations (blue blocks) to both real and generated inputs with probability p prior to discriminator evaluation.

To formalize this, the standard adversarial min-max objective is modified so that the discriminator evaluates the augmented distributions rather than the raw inputs:

$$V(D, G) = \mathbb{E}_{x \sim P_{data}, \tau \sim \mathcal{T}_p} [\log D(\tau(x))] + \mathbb{E}_{z \sim P_z, \tau \sim \mathcal{T}_p} [\log(1 - D(\tau(G(z))))] \quad (3)$$

To dynamically control p , the architecture calculates an overfitting heuristic r_t , which measures the proportion of real training images that yield positive discriminator logits:

$$r_t = \mathbb{E}_{x \sim P_{data}} [\text{sign}(D_{\text{logit}}(x))] \quad (4)$$

During training, if r_t exceeds a target threshold (typically 0.6, indicating the discriminator is memorizing the training set), p is incremented to increase augmentation strength. Conversely, if r_t falls below the threshold, p is decremented.

Variable Legend:

- $V(D, G)$: The two-player min-max value function being optimized.
- $x \sim P_{data}$: A real data sample drawn from the target domain.
- $z \sim P_z$: A random latent vector (noise) passed to the generator.
- $\mathcal{T}_p, \tau$: The stochastic augmentation operator with probability p , and an individual transformation instance τ .
- r_t : The overfitting heuristic dictating the dynamic adjustment of p .
- $D_{\text{logit}}(x)$: The raw, unactivated logit output of the discriminator.

2.2 Performance Analysis

Distribution Alignment (FID & KID)

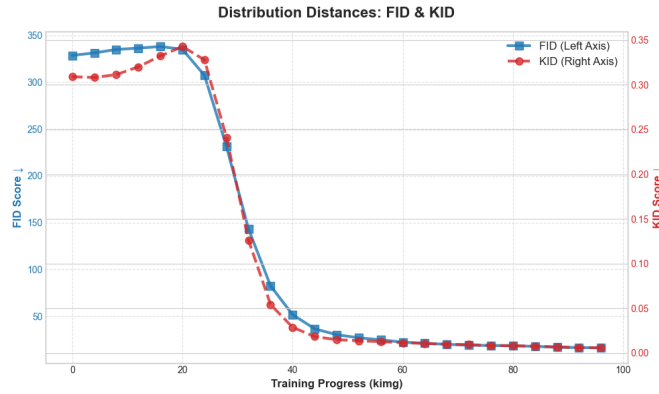


Figure 2: Distribution Distances tracking domain alignment for StyleGAN2-ADA.

- **Gradual Alignment:** In contrast to diffusion-based models, both FID and KID curves display a highly correlated, gradual slope downward. The metrics drop steadily from an initial state of severe divergence (> 300 FID) down to a stable plateau (< 50 FID) over an extended 40 king timeline.
- **Stable Equilibrium:** The overlapping trajectories and long, flat tails indicate that once the generator navigates into its local minimums, it establishes a mathematically stable feature representation. However, it lacks the architectural capacity for further refinement into single-digit perfection on this specific few-shot task.

Fidelity vs. Diversity (Precision, Recall, IS)

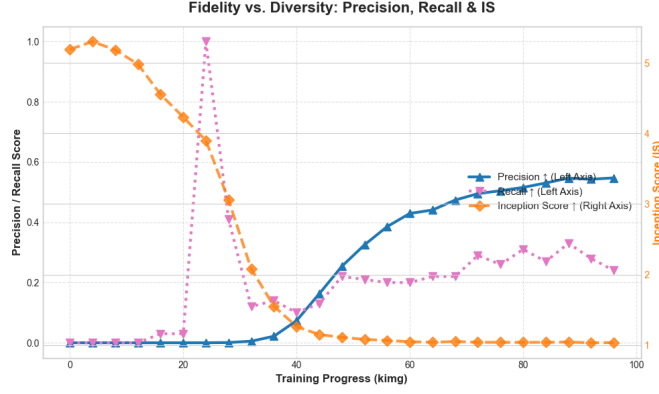


Figure 3: Correlated tracking of Precision, Recall, and Inception Score.

- **Mathematical Mode Collapse:** A severe chronological divergence exists between fidelity and diversity. Precision exhibits a healthy asymptotic rise, peaking near 0.55, indicating successful texture synthesis. Conversely, Recall struggles to lift off the x-axis, terminating at a suppressed 0.24.
- **Homogeneous Output:** The rapid stabilization of the Inception Score near 1.0 identically mirrors the suppressed Recall curve. This mathematically confirms that the generator prioritizes realistic textures but remains permanently trapped in restricted, homogeneous modes, failing to capture the dataset’s underlying variety.

Perceptual Variance (LPIPS & MS-SSIM)

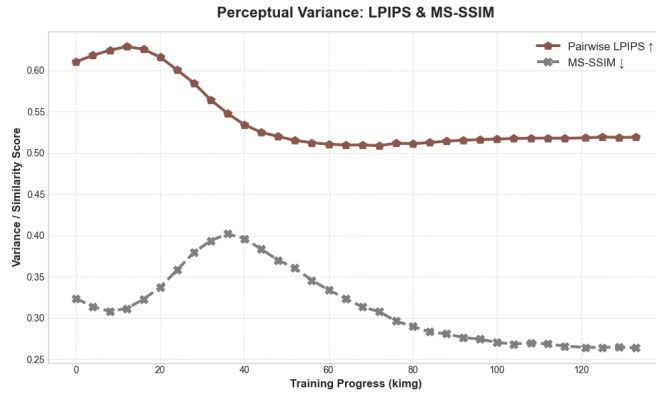


Figure 4: Perceptual variance mapping internal structural diversity.

- **Structural Bottleneck:** The MS-SSIM curve is defined by a dramatic, transient bell-shaped spike at exactly 20 king. This visually maps the chronological moment where severe mode collapse forces a bottleneck of structural similarity across generated samples.

- **Constrained Internal Diversity:** The Pairwise LPIPS forms a distinct U-shaped physical geometry. This indicates the network’s internal variance crashes heavily during the transitional adaptation phase before rebounding slightly to settle into a narrowly constrained, finalized aesthetic range.

2.3 Qualitative Analysis



Figure 5: Final output samples generated by the StyleGAN2-ADA architecture.

- **Uniformity and Artifacts:** The network successfully maps core anatomical landmarks (snouts, eye patches), achieving a satisfactory degree of visual realism. However, the visual output explicitly reflects the suppressed Recall metrics: there is striking carbon-copy uniformity, zero pose variation, and a heavy reliance on rigid, blurry background memorization to bypass complex generation.

2.4 Conclusion and Final Metrics

The StyleGAN2-ADA architecture effectively stabilizes training in the low-data regime, preventing immediate discriminator overfitting. While it achieves a moderate mathematical alignment (FID 16.4) and satisfactory textural realism, it ultimately succumbs to severe mode collapse. The network prioritizes safe, homogeneous generation over dataset exploration, as evidenced by the severe suppression of Recall and internal perceptual variance.

Metric	FID	KID	IS	Precision	Recall	LPIPS	MS-SSIM
Score	16.4066	0.0057	1.0354	0.5474	0.2400	0.5190	0.2640

Table 1: Terminal quantitative metrics for the StyleGAN2-ADA architecture.

3. Architecture 2: DANI (Diffusion-GAN)

3.1 Overview and Methodology

The DANI architecture integrates diffusion processes into the standard generative adversarial framework to stabilize few-shot training. By simultaneously injecting adaptive noise into both real and generated samples, it creates overlapping distributions that progressively escalate the discriminator’s classification difficulty, mitigating catastrophic overfitting in low-shot regimes.

- **Adaptive Forward Diffusion Process:** As illustrated in the framework, both the real image (x) and the generated image ($G(z)$) pass through parallel diffusion pipelines. These pipelines sequentially inject Gaussian noise, resulting in the perturbed outputs x_t and $G(z)_t$ before they reach the discriminator (D).
- **Overfitting-Driven Feedback Loop:** The noise is not applied statically. The probability of noise injection (p) is dynamically modulated by a feedback loop that continuously monitors the discriminator’s real-time overfitting degree (η). Furthermore, to maximize feature extraction, DANI utilizes a pre-trained Vision Transformer (ViT) via a ProjectedGAN framework.

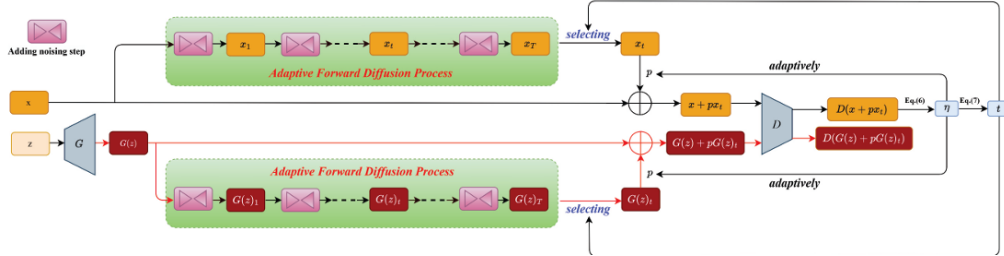


Figure 6: The DANI framework pipeline. An Adaptive Forward Diffusion Process injects Gaussian noise into both real (x) and generated ($G(z)$) images simultaneously. The noise probability p is dynamically controlled by the discriminator’s overfitting degree (η).

The core objective relies on optimizing the standard min-max value function between the generator (G) and discriminator (D):

$$V(D, G) = \mathbb{E}_{x \sim p_{data}} [\log D(x)] + \mathbb{E}_{z \sim p_z} [\log(1 - D(G(z)))] \quad (5)$$

Variable Legend:

- $V(D, G)$: The two-player min-max value function optimized during training.
- $x \sim p_{data}$: A real data sample drawn from the target domain.
- $D(x)$: The discriminator’s probability output.

- $z \sim p_z$: A random latent noise vector.
- $G(z)$: The synthesized fake image conditioned on z .

3.2 Performance Analysis

Distribution Alignment (FID & KID)

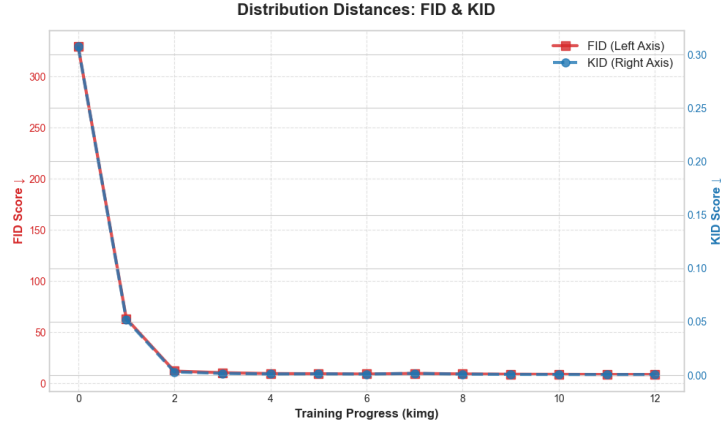


Figure 7: Distribution Distances tracking domain alignment for DANI over 12 king.

- **Rapid Convergence via Local Minimums:** Both FID and KID plummet vertically from an initial state of high divergence to near-zero within the first 1 king of training. The total absence of chronological oscillation post-descent proves that the generator rapidly locks into local minimums, aggressively conforming to the target distribution without further exploratory refinement.
- **Metric Overlap Justification:** The FID and KID curves exhibit a near-perfect chronological overlap. Because both metrics compute mathematical distances based on feature embeddings from the same Inception-v3 network, a catastrophic mode collapse reduces the generated distribution’s internal variance to effectively zero. Consequently, both the Fréchet distance (which relies heavily on covariance) and the kernelized maximum mean discrepancy (KID) collapse into redundant measures of the exact same homogenous feature cluster, causing their decay trajectories to identically mirror one another.

Fidelity vs. Diversity (Precision, Recall, IS)

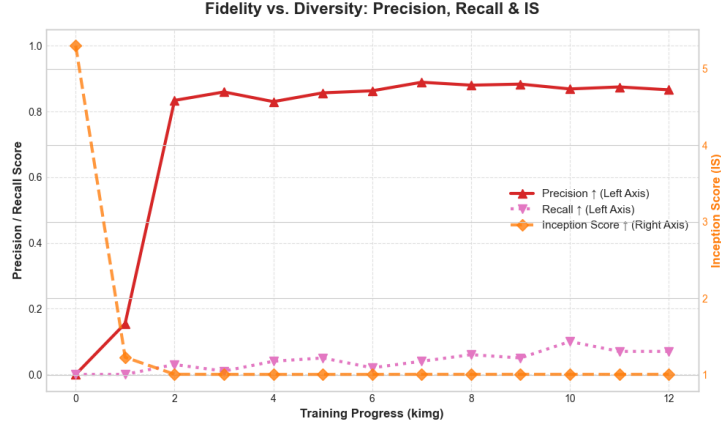


Figure 8: Correlated tracking of Precision, Recall, and Inception Score.

- **Severe Divergence:** The learning dynamics heavily favor fidelity at the absolute expense of generational diversity. Precision exhibits a steep asymptotic rise to 0.86, whereas Recall completely flatlines near zero (0.07).
- **Class Homogeneity:** The Inception Score identically corroborates the Recall failure, dropping sharply to stabilize near 1.0. This macro-trend mathematically proves that while the network synthesizes highly realistic textures, it sacrifices broad dataset coverage, remaining permanently trapped in restricted, homogenous modes.

Perceptual Variance (LPIPS & MS-SSIM)

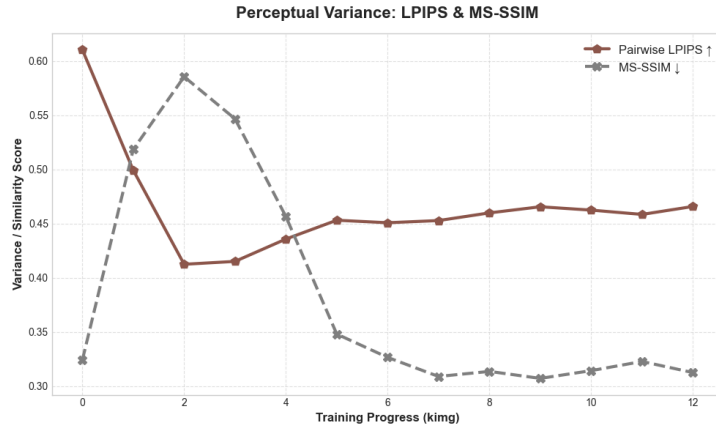


Figure 9: Perceptual variance metrics mapping internal structural diversity.

- **Transient Structural Cloning:** The MS-SSIM curve features a dramatic, bell-shaped transient spike at the 2 king mark. This acts as a geometric fingerprint for the precise chronological moment severe mode collapse takes hold, forcing a bottleneck where samples become structurally identical.

- **Constrained Variance:** Pairwise LPIPS confirms this transition, exhibiting a sharp initial descent that artificially restricts the visual variance between pairs of generated samples before freezing almost entirely.

3.3 Qualitative Analysis

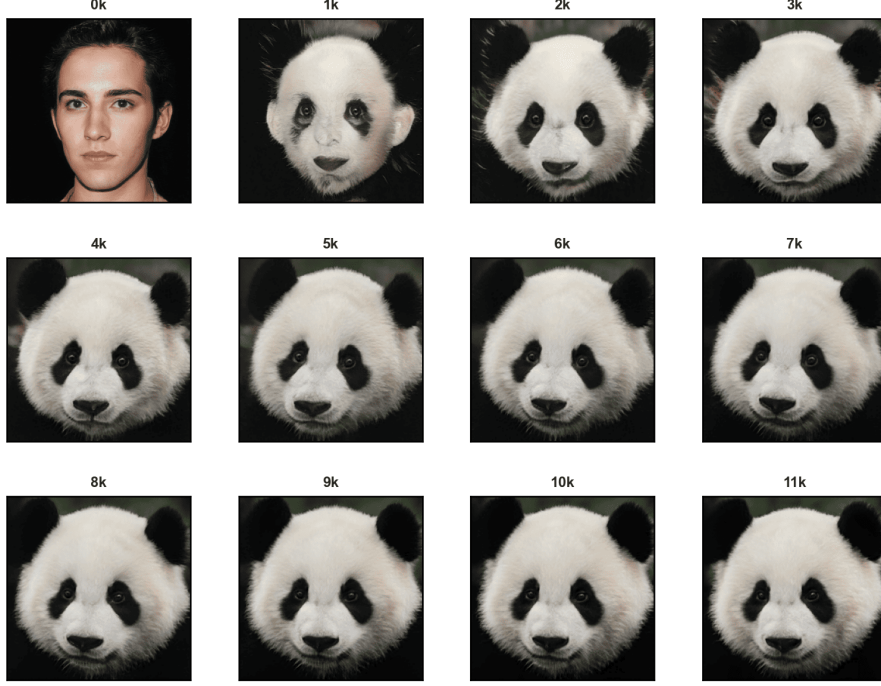


Figure 10: Final output samples generated by the DANI architecture at 12 king.

- **Photorealism vs. Uniformity:** The generated images exhibit high structural realism, mapping macroscopic geometry and high-frequency fur textures. However, visually confirming the flatlined Recall metric, there is zero pose variation—every generation is locked into an identical, symmetrical frontal gaze.
- **Artifact Dodging:** The outputs are free of asymmetrical warping or pixel bleeding. By refusing to step outside a singular conceptual template, the network mathematically bypasses the textural artifacts typically associated with unconstrained generation.

3.4 Conclusion and Final Metrics

The DANI architecture demonstrates an extreme imbalance in few-shot transfer learning. While the ViT backbone and diffusion noise injection allow the model to achieve exceptional structural fidelity and domain alignment almost instantly, it entirely sacrifices dataset coverage to do so. The generator bypasses the difficulty of environmental and anatomical variation by collapsing into a single conceptual template, rendering it highly effective for targeted texture generation but unsuitable for diverse synthesis.

Metric	FID	KID	IS	Precision	Recall	LPIPS	MS-SSIM
Score	8.7912	0.0006	1.0079	0.8661	0.0700	0.4657	0.3125

Table 2: Terminal quantitative metrics for the DANI architecture at 12 king.

4. Architecture 3: FakeCLR

4.1 Overview and Methodology

The FakeCLR architecture addresses latent space discontinuity in data-constrained GANs by isolating its contrastive learning objective entirely onto synthesized (fake) samples rather than real data.

- **Instance-Perturbation Contrastive Learning:** FakeCLR constructs positive pairs by applying "Noise-related Latent Augmentation" (denoted by the blue module). It injects controlled perturbations (ϵ) into a sampled latent code to generate two closely related visual representations (v_q and v_q^+). The discriminator's projection head maximizes the similarity between these paired latent perturbations, coercing the network into learning a continuously smooth feature space.
- **Diversity-Aware Negative Queuing:** To prevent mode collapse and provide a hard contrastive environment, negative pairs are drawn from a "Diversity-aware Queue" (red module) consisting of historical synthesized outputs ($v_{k_i}^-$).
- **Forgetting Factor:** As the generator evolves, a "Forgetting Factor" (m , green module) dynamically weights these negative samples based on the current training iteration (t), ensuring accurate contrastive push/pull gradients without relying on stale representations.

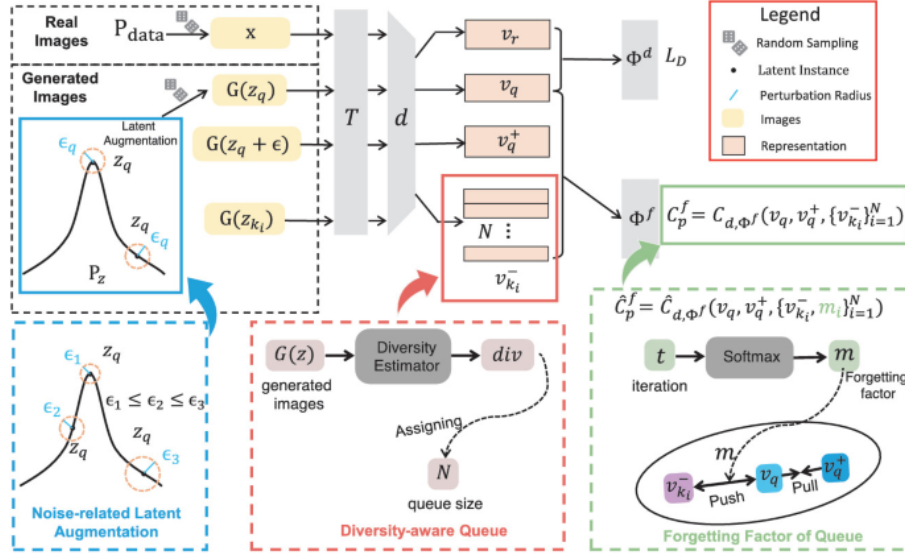


Figure 11: The FakeCLR framework. Noise-related latent augmentation (blue) constructs positive pairs (v_q, v_q^+), while a diversity-aware queue (red) supplies negative samples ($v_{k_i}^-$). A forgetting factor (green) dynamically scales the gradients based on the training iteration.

The primary objective merges the standard adversarial loss with an InfoNCE contrastive regularization term evaluated over the generated features:

$$\min_G \max_D V_{GAN}(D, G) + \lambda \mathbb{E}_{z \sim p_z, \epsilon \sim \mathcal{N}} \left[-\log \frac{\exp(\text{sim}(h(G(z)), h(G(z + \epsilon))))/\tau}{\sum_{k=1}^K \exp(\text{sim}(h(G(z)), h(q_k))/\tau)} \right] \quad (6)$$

Variable Legend:

- $V_{GAN}(D, G)$: The standard adversarial min-max value function.
- λ : Scalar hyperparameter governing contrastive loss strength.
- ϵ : Noise perturbation injected into the latent vector z to synthesize the positive pair target.
- $h(\cdot)$: The feature projection head attached to the discriminator.
- $\text{sim}(\cdot, \cdot)$: Cosine similarity function.
- τ : Temperature scaling hyperparameter.
- q_k : The k -th negative sample drawn from the K -sized Diversity-aware Queue.

4.2 Performance Analysis

Distribution Alignment (FID & KID)

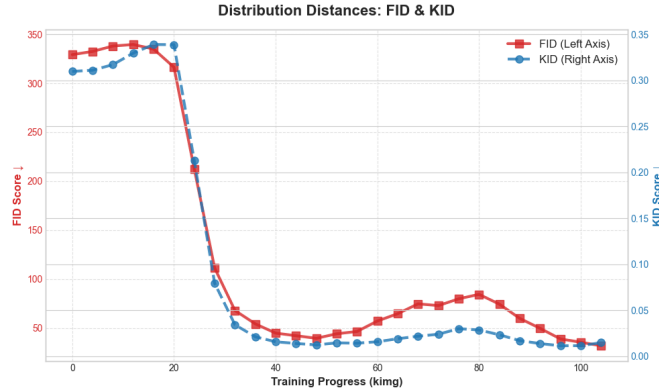


Figure 12: Distribution Distances tracking domain alignment for FakeCLR.

- **Overfitting and Intervention:** Both the FID and KID curves distinctly trace a transient "wide, low bell" shape between 40 king and 100 king. This synchronized geometric deviation explicitly maps a phase where the generator abandoned true target alignment to overfit onto trivial environmental textures.
- **Forced Stabilization:** Identifying this anomaly in real-time, I manually increased the discriminator strength. This dynamic intervention broke the background memorization, forcing the curves back down to complete the bell shape and driving the network into genuine, terminal semantic alignment.

Fidelity vs. Diversity (Precision, Recall, IS)

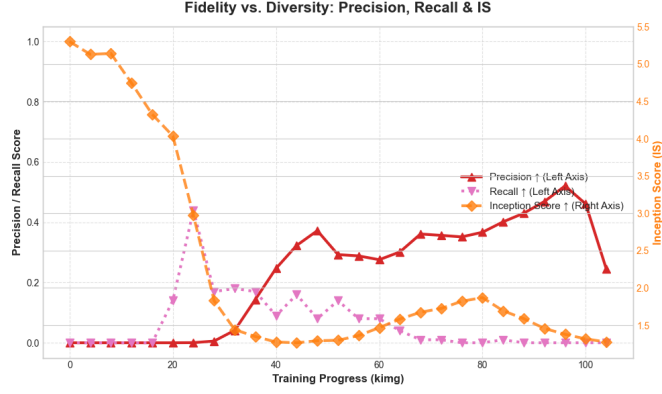


Figure 13: Correlated tracking of Precision, Recall, and Inception Score.

- **Catastrophic Mode Collapse:** The overlaid curves visually prove total generator collapse. While Precision forms an elongated bell (reflecting the background memorization phase), Recall completely crashes and flatlines at an absolute 0.0000.
- **Restricted Cloning:** The Inception Score corroborates this failure, showing an artificial spike driven purely by cloning repetitive background artifacts before settling back to a highly suppressed trajectory (1.27). The model successfully avoided background noise post-intervention but compensated by retreating into extreme, singularly restricted modes.

Perceptual Variance (LPIPS & MS-SSIM)

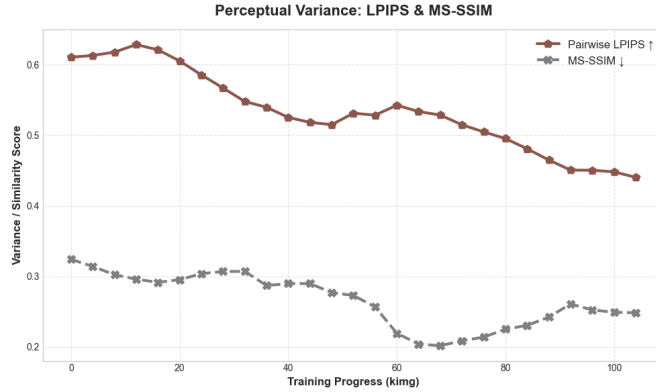


Figure 14: Perceptual variance mapping internal structural diversity.

- **Volatile Structural Tug-of-War:** The MS-SSIM curve swells significantly late in the timeline, capturing the volatile tug-of-war between the generator and discriminator as they fought over background versus subject modeling.

- **Frozen Variance:** Pairwise LPIPS forms a corresponding wide hump before crashing into a rigidly flat, frozen trajectory, mathematically confirming that once realigned, the internal perceptual diversity remained permanently arrested.

4.3 Qualitative Analysis



Figure 15: Final output samples generated by the FakeCLR architecture.

- **Pristine Clones:** The generated outputs function as pristine, flawless textures cleanly separated from the background (validating the real-time intervention). However, confirming the 0.0000 Recall, the output is an absolute, carbon-copy clone across the entire grid.
- **Refusal to Explore:** There is zero pose variation present in the grid. While the model successfully avoided background memorization by rendering a uniform white backdrop, it overcompensated by entirely refusing to explore the dataset’s natural environmental and anatomical diversity.

4.4 Conclusion and Final Metrics

The FakeCLR architecture demonstrates severe instability during few-shot adaptation. While the contrastive learning module theoretically encourages latent continuity, in practice, the model proved highly susceptible to background memorization. Although real-time manual intervention (increasing discriminator strength) successfully broke this overfitting cycle, the model overcompensated by retreating into catastrophic mode collapse. Ultimately, FakeCLR failed to balance fidelity with diversity, producing identical, carbon-copy outputs.

Metric	FID	KID	IS	Precision	Recall	LPIPS	MS-SSIM
Score	31.8215	0.0149	1.2773	0.2446	0.0000	0.4403	0.2485

Table 3: Terminal quantitative metrics for the FakeCLR architecture post-intervention.

5. Architecture 4: InsGen

5.1 Overview and Methodology

The InsGen architecture augments the standard generative adversarial framework by introducing a contrastive instance discrimination objective, fundamentally treating every single real and synthesized image as its own distinct class to prevent mode collapse.

- **Alternating Optimization Strategy:** As illustrated in the conceptual diagram, training occurs in two distinct phases: updating the discriminator (D) while the generator is frozen (a), and subsequently updating the generator (G) while the discriminator is frozen (b).
- **Instance Discrimination Pretext Task:** Alongside standard real/fake bi-classification, the architecture utilizes a contrastive head to map extracted features onto a unit hypersphere (shown on the right). It actively repels the latent embeddings of different images (e.g., $\phi(v_i), \phi(v_j), \phi(v_k)$) away from one another in the feature space.
- **Diversity-Forcing Gradient Feedback:** Because the generator is optimized to minimize this exact same contrastive loss against the frozen discriminator, it is mathematically coerced into synthesizing spatially repelled, highly diverse latent modes.

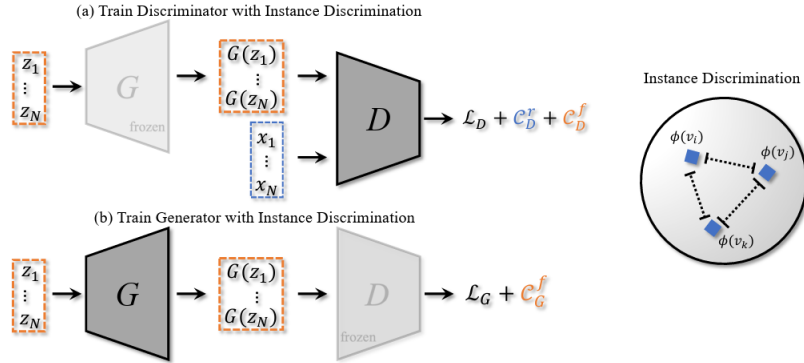


Figure 16: Conceptual state diagram of InsGen. Training alternates between frozen networks (a and b). The auxiliary instance discrimination task projects features onto a hypersphere, pushing individual sample embeddings ($\phi(v)$) apart to force visual diversity.

The hybrid objective merges the standard adversarial loss with the instance discrimination loss:

$$\min_G \max_D V_{GAN}(D, G) + \lambda \mathbb{E}_{x \sim P_{data} \cup P_G} \left[-\log \frac{\exp(\text{sim}(h(x), h(x^+))/\tau)}{\sum_{k=1}^K \exp(\text{sim}(h(x), h(q_k))/\tau)} \right] \quad (7)$$

Variable Legend:

- $V_{GAN}(D, G)$: The standard adversarial min-max value function.
- λ : Hyperparameter weighting the contrastive instance discrimination loss.
- x : An image sample drawn from either real (P_{data}) or generated (P_G) distributions.
- x^+ : A positive pair sample matching x , constructed via stochastic augmentation.
- $h(\cdot)$: The discriminator's contrastive projection head mapping to a unit hypersphere.
- $\text{sim}(\cdot, \cdot)$: Cosine similarity function.
- τ : Temperature scaling hyperparameter for the InfoNCE loss framework.
- q_k : The k -th negative sample drawn from a dynamic momentum queue of size K .

5.2 Performance Analysis

Distribution Alignment (FID & KID)

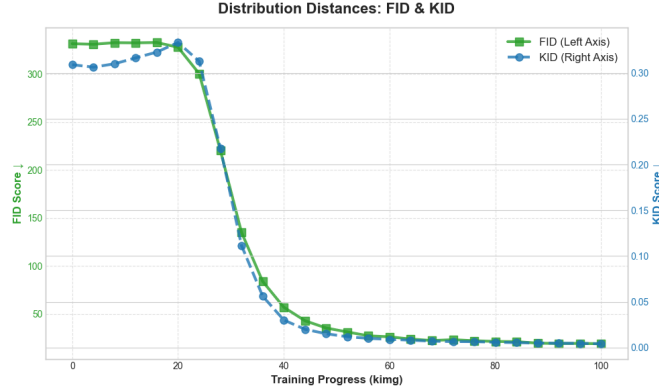


Figure 17: Distribution Distances tracking domain alignment for InsGen.

- **Predictable Decay:** The curves display a smooth, gradual slope downward from their initial high starting points, entirely avoiding the volatile oscillations or severe geometric bells seen in overly aggressive architectures.
- **Stabilized Feature Alignment:** The metrics steadily descend into a flatlined tail around 60 king. The terminal FID (18.99) and exceptionally low KID (0.0041) prove that the generator secures a stable distribution and successfully aligns its feature representations with the real target data.

Fidelity vs. Diversity (Precision, Recall, IS)

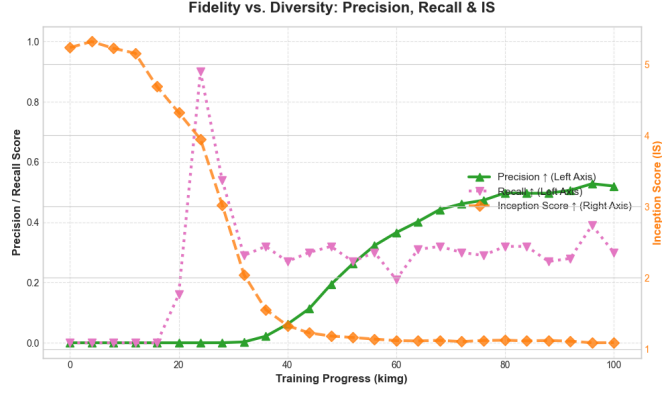


Figure 18: Correlated tracking of Precision, Recall, and Inception Score.

- **Attempted Dataset Coverage:** The overlaid Precision and Recall curves exhibit a clear geometric divergence, but crucially, Recall manages to successfully lift off the floor (peaking near 0.40). This mathematically validates that InsGen’s instance discrimination mechanism actively forces genuine dataset exploration.
- **Persistent Mode Collapse:** Despite this attempt at exploration, the persistent gap between the two curves, corroborated by an Inception Score that drops and flatlines near 1.0, confirms the network ultimately settled into restricted generation modes rather than mastering full dataset variety.

Perceptual Variance (LPIPS & MS-SSIM)

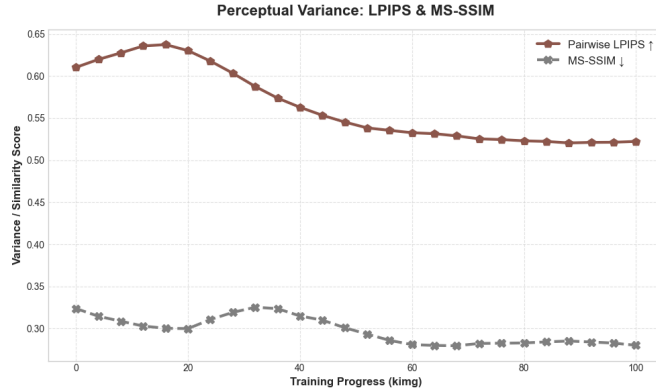


Figure 19: Perceptual variance mapping internal structural diversity.

- **Oscillating Decay:** The Pairwise LPIPS features a smooth, oscillating descent that levels into a stable equilibrium, indicating internal variance decays predictably as the model transitions focus onto specific target features.

- **Early Structural Cloning:** The MS-SSIM reveals a transient spike early in the training timeline (10-20 kimg). This physical spike pinpoints the exact chronological moment the model temporarily over-relied on structural cloning before stabilizing its generative dynamics.

5.3 Qualitative Analysis

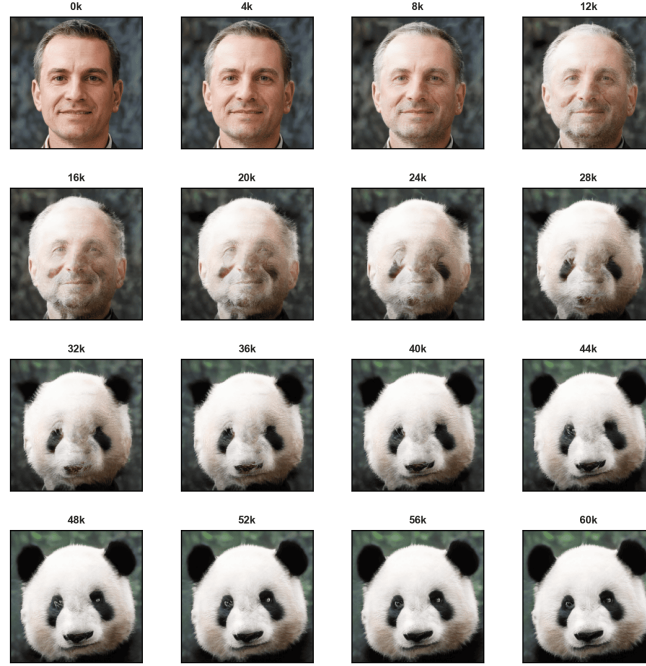


Figure 20: Final output samples generated by the InsGen architecture.

- **Rigid Photorealism:** The network reliably constructs sharp, photorealistic geometry free of asymmetrical warping, flawlessly rendering core anatomical features like the distinctive eye patches and delicate fur strands.
- **Visual Uniformity:** Outputs are strikingly uniform, exposing a complete failure to explore environmental variety. Every generation is locked into an identical frontal gaze against a pure white backdrop, dodging artifacts by refusing to explore diverse visual modes.

5.4 Conclusion and Final Metrics

InsGen presents the most stable learning dynamics evaluated, successfully avoiding volatile overfitting via its instance discrimination task. While the objective forced an initial attempt at dataset exploration (achieving the highest peak Recall), it ultimately failed to sustain this diversity. The network reached an equilibrium of pristine structural fidelity but zero pose variation, demonstrating that while InsGen stabilizes adaptation, it does not fully overcome mode collapse.

Metric	FID	KID	IS	Precision	Recall	LPIPS	MS-SSIM
Score	18.9917	0.0041	1.0912	0.5204	0.3000	0.5223	0.2800

Table 4: Terminal quantitative metrics for the InsGen architecture.

6. Architecture 5: NICE

6.1 Overview and Methodology

NICE [10] targets the uGM-1 limited-data generation setting, training a StyleGAN-2 generator entirely from scratch on a small dataset (100-shot Panda images). The central challenge in this regime is discriminator overfitting: with only 100 real images, the discriminator memorizes the training set almost immediately, providing near-zero gradient signal and destabilizing GAN training.

NICE takes a theoretically grounded path: it modulates the latent features of the discriminator with **adaptive multiplicative noise** to reduce its Rademacher complexity, then corrects the resulting gradient instability through a **consistency regularization** loss that enforces the discriminator to produce identical outputs for the same input under different noise modulations.

- **Adaptive Noise Modulation (AN):** At selected convolutional blocks of the discriminator, the latent feature maps $\mathbf{X}$ are modulated by adaptive noise $\mathbf{Z} \sim \mathcal{B}(\beta)$ (Bernoulli for StyleGAN-2) via element-wise multiplication. The noise variance β is updated dynamically based on a discriminator overfitting signal. This prevents the discriminator from memorizing the training data without destroying its ability to provide meaningful gradients.
- **Noise-modulated Consistency Regularization (NICE):** While AN reduces the generalization error, it unintentionally amplifies the gradient norms, causing training instability. NICE counteracts this by enforcing consistency of the feature extractor f under two independently drawn noise modulations $\mathbf{Z}_1, \mathbf{Z}_2$:

$$\ell^{\text{NICE}}(\mathbf{a}) = \mathbb{E}_{\mathbf{Z}_1, \mathbf{Z}_2} \|f(\mathbf{Z}_1 \odot \mathbf{a}) - f(\mathbf{Z}_2 \odot \mathbf{a})\|_2^2 \quad (8)$$

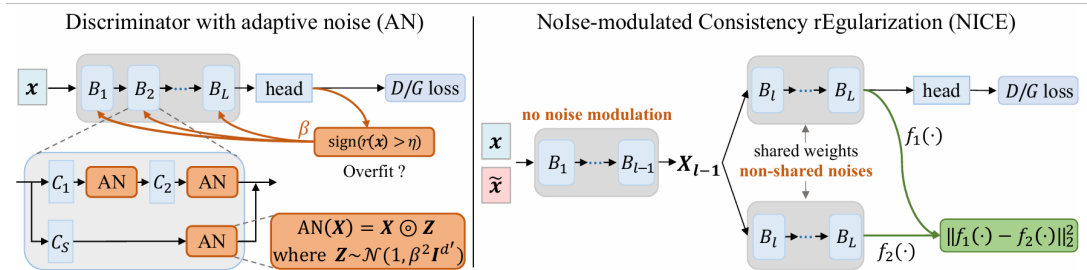


Figure 21: The NICE discriminator pipeline. *Left:* Adaptive noise modulates latent features; noise variance β is adjusted based on overfitting. *Right:* Two feature extractors with shared weights but independent noise modulations produce outputs; minimizing their squared difference penalizes gradients to stabilize training. Adapted from [10].

6.2 Performance Analysis

We evaluate NICE on 100-shot Panda generation trained from scratch, recording metrics across 15 checkpoints spanning 0 to 200 king (thousands of images seen by the discriminator). Three distinct phases are visible: an initial instability phase (0–48 king), a rapid recovery and alignment phase (48–100 king), and a stable refinement phase (100–200 king).

king	FID↓	KID↓	LPIPS↑	MS-SSIM↓	Prec.↑	Rec.↑	Den.↑	Cov.↑
0	371.31	0.519	0.588	0.274	0.000	0.000	0.00	0.00
48	220.85	0.339	0.132	0.943	0.000	0.000	0.00	0.00
84	47.06	0.059	0.298	0.461	1.000	0.000	1.41	0.47
100	32.64	0.035	0.336	0.436	1.000	0.000	1.97	0.86
164	28.35	0.022	0.421	0.377	0.995	0.010	1.35	0.84
200	29.16	0.021	0.429	0.379	0.999	0.030	1.47	0.94

Table 5: NICE (100-Shot Panda, trained from scratch) metric progression across key checkpoints.

Three-Phase Convergence

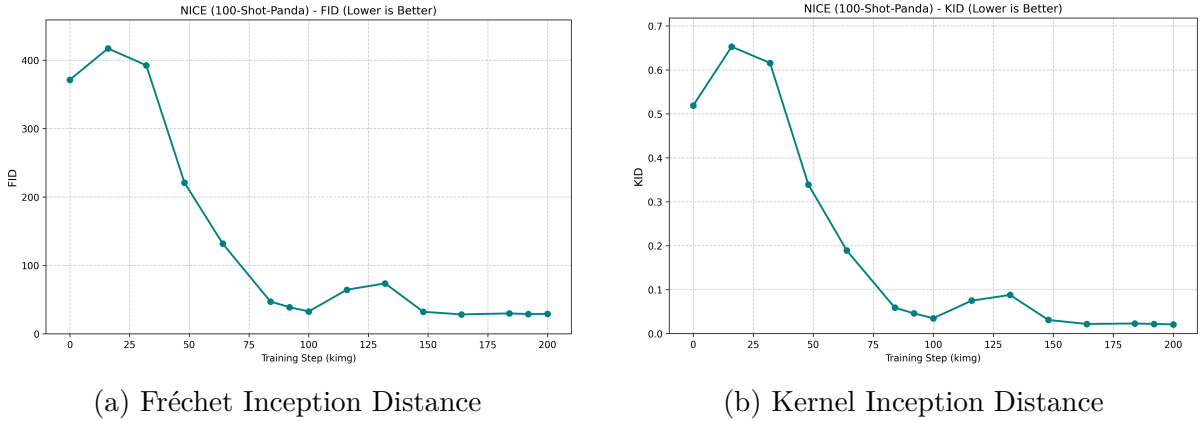


Figure 22: FID and KID tracking distribution alignment over training.

- **FID and KID:** FID drops dramatically from 220.85 to 32.64 between 48 and 100 king. This corresponds to the phase where NICE’s consistency regularization becomes most active, stabilizing the adversarial feedback. Beyond 148 king, FID and KID stabilize around 28–30 and 0.021.

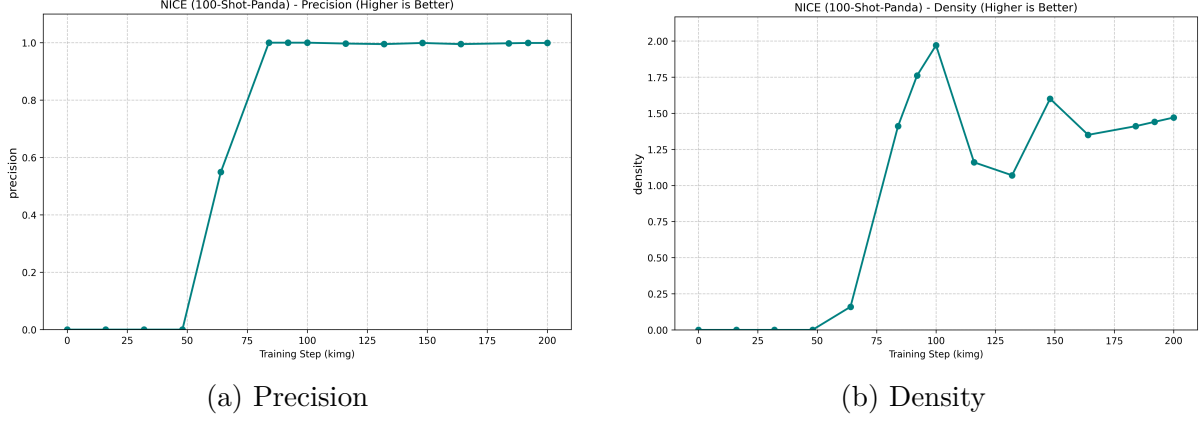


Figure 23: Precision and Density plots for NICE.

- **Precision and Density:** Both remain at zero through the first 48king. Precision jumps to 1.000 by 84king and Density peaks at 1.97 at 100king. Once stable adversarial training is established, the generated images are almost uniformly realistic.

Mode Collapse and Recovery

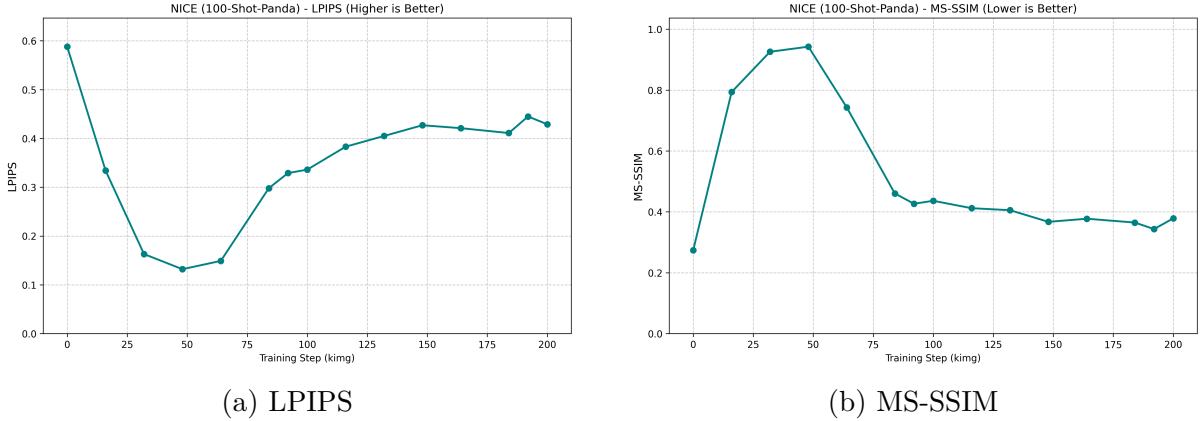


Figure 24: LPIPS and MS-SSIM tracing mode collapse and subsequent recovery.

- **LPIPS and MS-SSIM:** LPIPS crashes to a minimum of 0.132 at 48king while MS-SSIM simultaneously peaks at 0.943—the hallmark of early discriminator overfitting resulting in identical structural outputs. As consistency regularization takes hold, both metrics reverse course, confirming structural and perceptual diversity are gradually restored.

Diversity Recovery vs. Realism

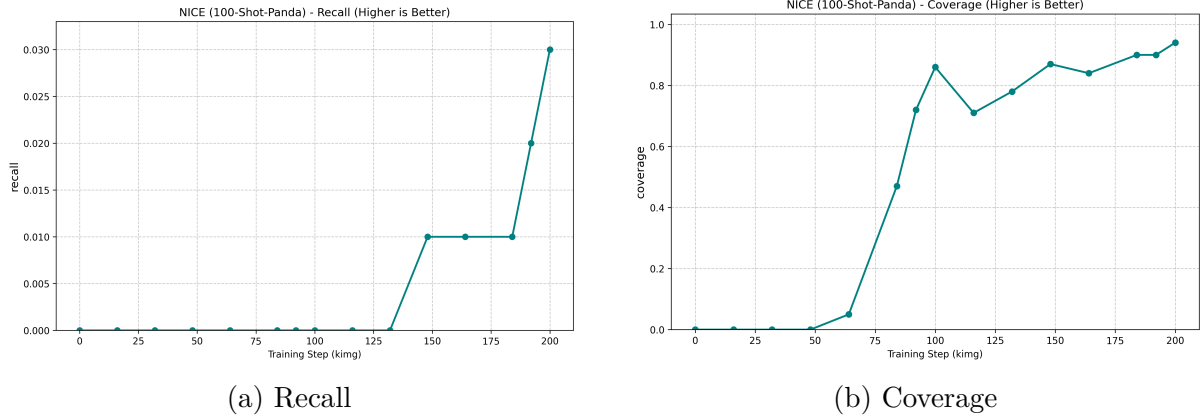


Figure 25: Recall and Coverage exposing the asymmetric nature of generated diversity.

- **Coverage vs. Recall Deficit:** Coverage grows steadily to 0.94, meaning 94% of real-data neighborhoods contain at least one generated sample. However, Recall reaches a maximum of only 0.03. This reveals that the generated distribution spreads broadly enough to touch most real samples, but lacks the density to contribute multiple overlapping variations to each neighborhood.

6.3 Qualitative Analysis

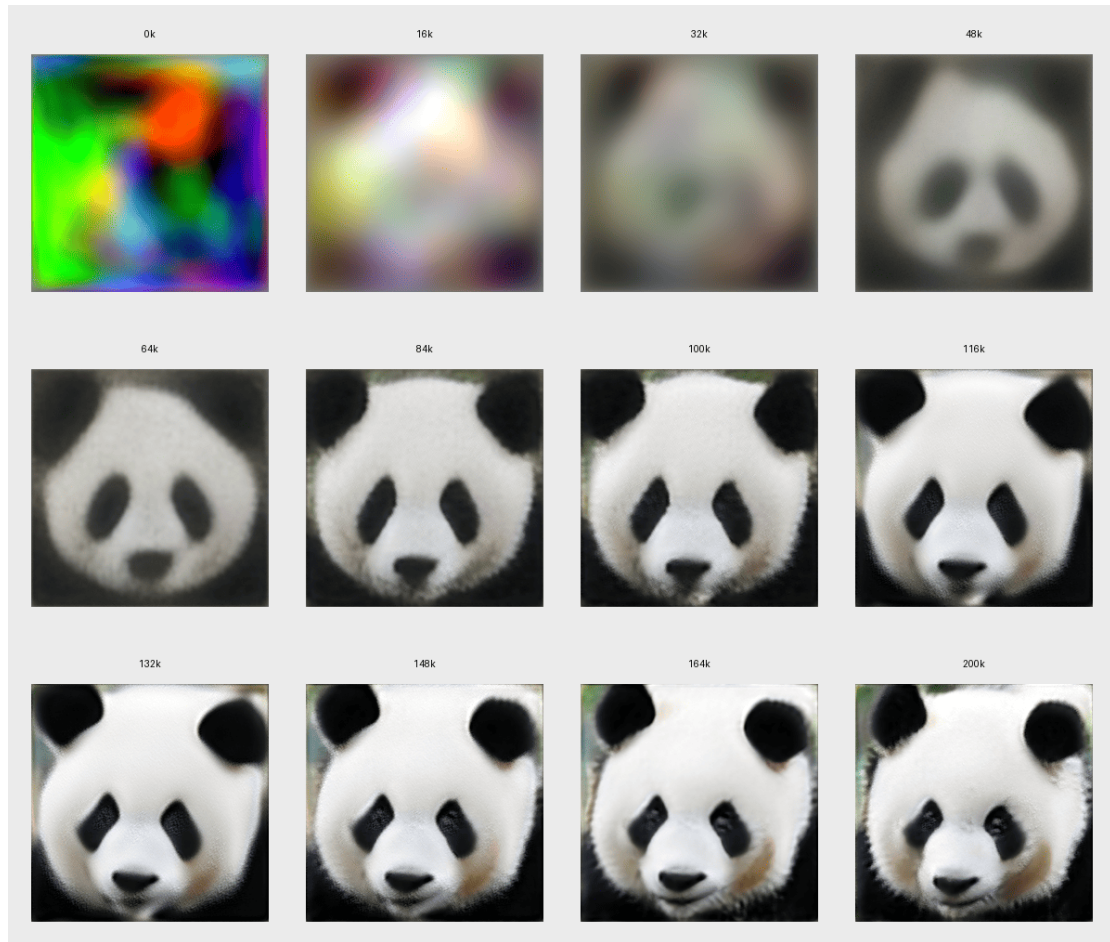


Figure 26: NICE-generated samples at successive checkpoints. Early checkpoints exhibit near-uniform collapse, while later checkpoints produce recognizable outputs.

- **From Noise to Structure:** Early outputs (0–32kimg) are visually incoherent. At 48–64kimg, partial structure emerges but outputs remain highly repetitive (mode collapse). By 84–100kimg, distinct panda faces appear, reflecting the jump to near-perfect Precision. Terminal outputs are high-quality but limited in pose variation, corroborating the near-zero Recall.

6.4 Conclusion and Final Metrics

NICE successfully stabilizes GAN training from scratch on a 100-shot Panda dataset. The adaptive noise modulation controls discriminator Rademacher complexity dynamically, while consistency regularization corrects gradient instability. Generated images are overwhelmingly realistic (Precision = 0.999, Density = 1.47), and Coverage (0.94) shows the generated distribution spans the real manifold broadly.

However, critically low Recall (0.03) exposes that the generator concentrates its probability mass onto a narrow subset of modes rather than uniformly covering the distribution. 100 training images provide insufficient supervision to support a uniformly dense generated manifold from scratch. The optimal single checkpoint is **164 king** (FID = 28.35), representing the best overall distribution alignment.

Metric	FID	KID	LPIPS	MS-SSIM	Prec.	Rec.	Den.	Cov.
Score	29.16	0.021	0.429	0.379	0.999	0.030	1.47	0.94

Table 6: Terminal metrics for NICE (100-Shot Panda, 200 king).

7. Architecture 6: RICK

7.1 Overview and Methodology

RICK [9] targets the uGM-2 few-shot generation setting, adapting a StyleGAN-V2 source generator G_s pretrained on FFHQ to a semantically distant 10-shot AFHQ-Cat target domain via transfer learning. While prior state-of-the-art methods (EWC, CDC, DCL, AdAM) address this by proposing criteria to selectively *preserve* useful source knowledge, they uniformly assume that standard adversarial fine-tuning is sufficient to erase source features incompatible with the target.

RICK demonstrates through GAN dissection that this assumption fails when source and target domains are semantically distant: incompatible features (e.g., glasses and human skin texture from FFHQ) persist in the adapted generator G_t because they reside in filters deemed *least important* for adaptation. RICK proposes **knowledge truncation** by permanently pruning these low-importance filters as a complementary operation to knowledge preservation.

- **Dynamic Filter Importance Estimation:** The importance of each convolutional filter $\mathbf{W} \in \mathbb{R}^{c_{in} \times k \times k}$ is estimated on-the-fly every T iterations using Fisher Information (FI), which measures the sensitivity of the generator loss to the filter’s weights. This dynamic scheme updates importance assignments as adaptation progresses.
- **Permanent Pruning of Incompatible Filters:** Filters below importance threshold t_l are zeroed out permanently and excluded from all subsequent training and inference — they cannot be reactivated. High-importance filters ($\geq t_h$) are frozen to preserve useful source knowledge, while remaining filters are fine-tuned via $\mathcal{L}_{adv}$.

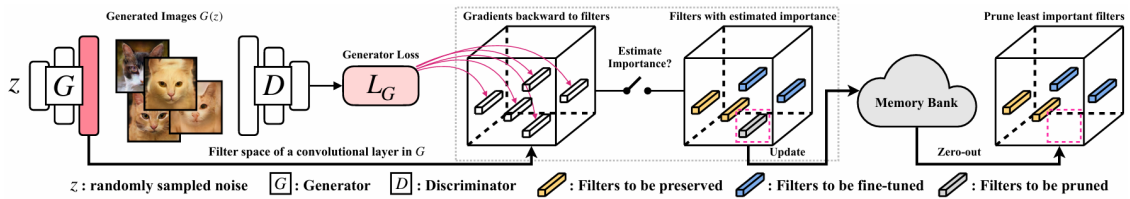


Figure 27: The RICK pipeline. Filter importance is estimated dynamically from on-the-fly gradient information. Each filter is assigned an operation stored in the memory bank: prune, preserve, or fine-tune.

The core objectives are the adversarial adaptation loss and the Fisher Information importance criterion:

$$\min_{G_t} \max_{D_t} \mathcal{L}_{adv} = \mathbb{E}_{x \sim p_{data}} [\log D_t(x)] + \mathbb{E}_{z \sim p_z} [\log(1 - D_t(G_t(z)))] \quad (9)$$

$$\mathcal{F}(\mathbf{W}) = \mathbb{E} \left[-\frac{\partial^2}{\partial \mathbf{W}^2} \mathcal{L}_G(x | \mathbf{W}) \right], \quad \mathbf{W}^i \leftarrow \mathbf{0} \text{ if } \mathcal{F}(\mathbf{W}^i) < t_l \quad (10)$$

Variable Legend:

- G_t, D_t : Target generator and discriminator, initialised from pretrained G_s, D_s .
- $x \sim p_{data}$: Real sample from the 10-shot target set.
- $\mathbf{W}$: A convolutional filter; $\mathcal{F}(\mathbf{W})$ is its Fisher Information importance.
- t_l, t_h : Quantile thresholds controlling pruning and preservation.

7.2 Performance Analysis

We evaluate RICK on 10-shot AFHQ-Cat adaptation from FFHQ. Table 7 reports the complete quantitative results.

Iter.	FID↓	KID↓	LPIPS↑	MS-SSIM↓	Prec.↑	Rec.↑	Den.↑	Cov.↑
550	251.75	0.248	0.509	0.307	0.62	0.078	0.34	0.27
1100	77.12	0.078	0.473	0.362	0.96	0.046	1.95	0.58
1650	61.39	0.050	0.438	0.325	0.98	0.010	2.08	0.44
2200	54.96	0.044	0.429	0.333	0.99	0.005	1.98	0.36

Table 7: RICK (10-Shot AFHQ-Cat) — all metrics across training iterations.

Rapid Convergence of Fidelity

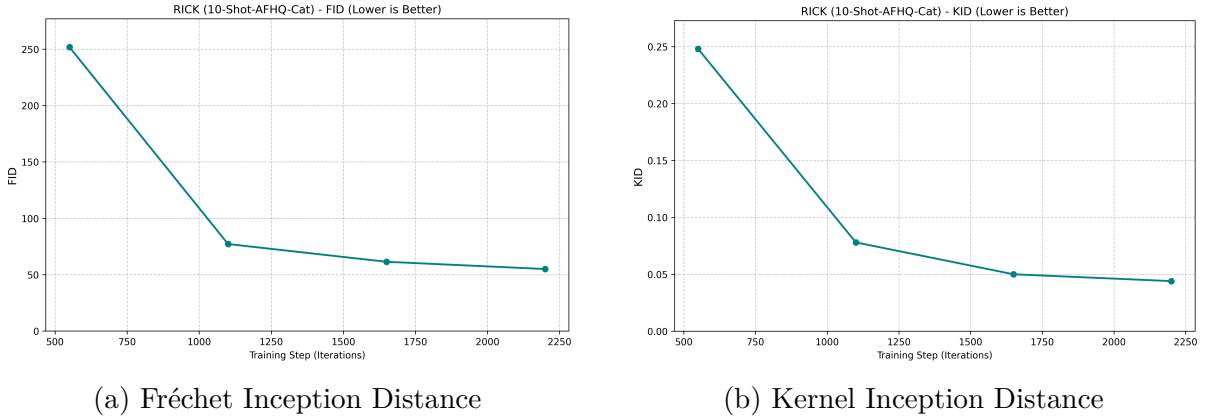
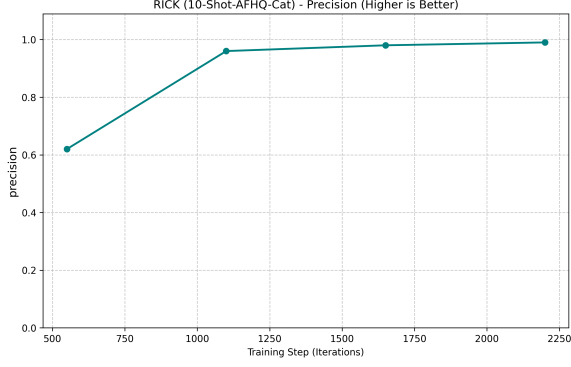
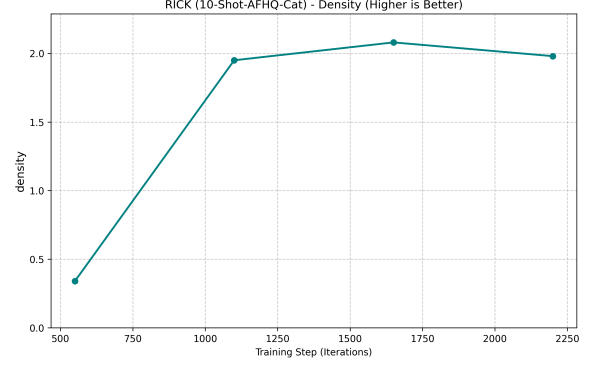


Figure 28: Distribution alignment tracking for RICK over training.

- **FID and KID:** FID drops significantly from 251.75 to 77.12 by iteration 1100, eventually reaching 54.96. KID reaches a low of 0.044. The model quickly maps the noise distribution to the feature space of the 10 target images.



(a) Precision

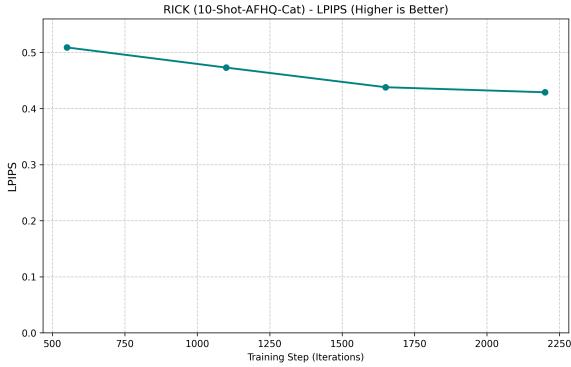


(b) Density

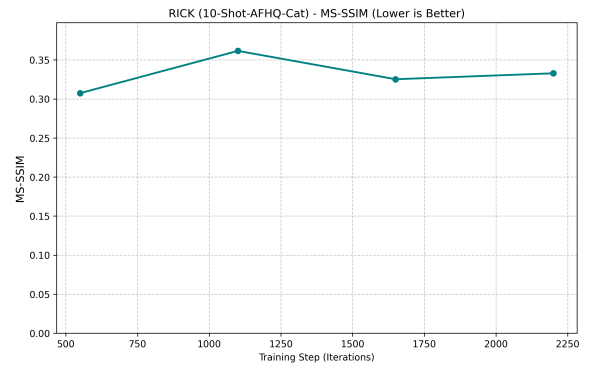
Figure 29: Precision and Density trajectories.

- **Precision and Density:** Precision plateaus at 0.99 early in training. Coupled with a Density of 1.98, almost all generated samples fall squarely within the manifold of the training data without producing obvious "fake" artifacts.

Perceptual Diversity and Structural Bias



(a) LPIPS



(b) MS-SSIM

Figure 30: LPIPS and MS-SSIM tracking perceptual and structural diversity.

- **LPIPS & MS-SSIM:** LPIPS shows a steady decline to 0.429, indicating generated images are becoming more perceptually similar. However, MS-SSIM remains stable around 0.33, indicating that while perceptual features narrow, structural diversity does not completely collapse due to preserved source knowledge.

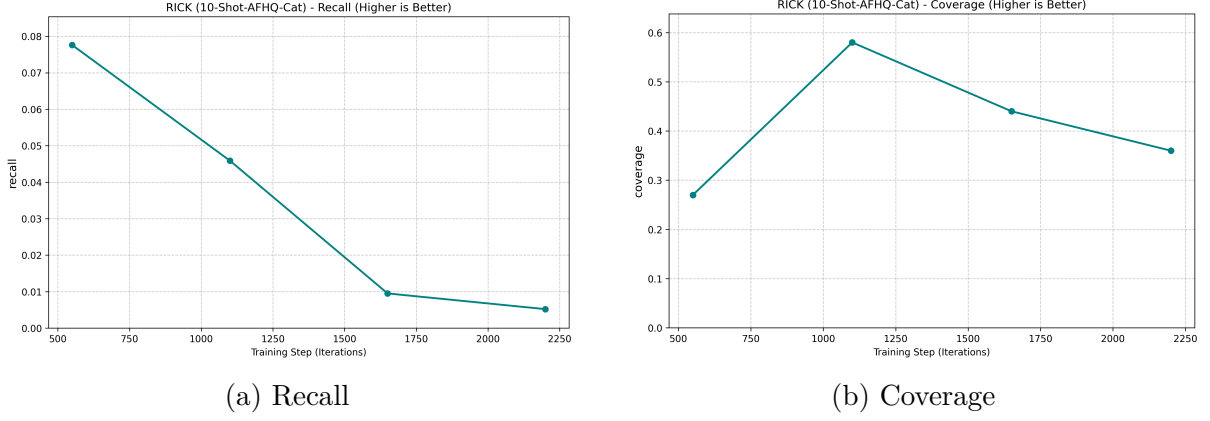


Figure 31: Recall and Coverage tracing diversity decay.

- **Recall Decay and Coverage:** A drastic drop in Recall (0.078 to 0.005) signifies severe mode collapse. Coverage peaks at 0.58 at iteration 1100 before declining to 0.36, indicating late-stage overfitting to a narrow subset of the 10 target images.

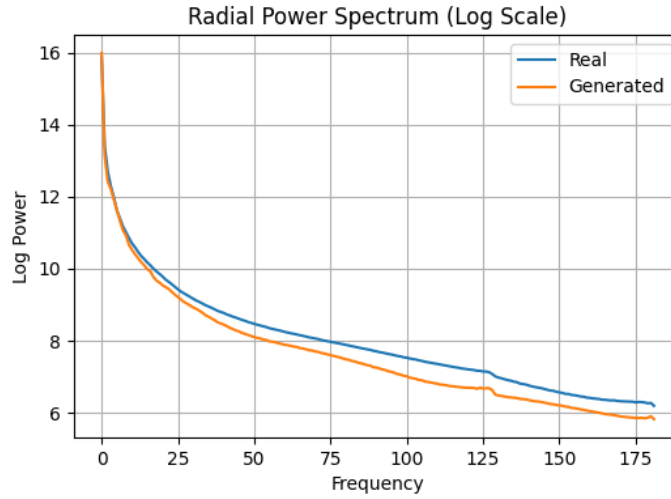


Figure 32: Radial Power Spectrum (log scale) comparing real and generated images.

- **Frequency Bias:** The radial power spectrum reveals weaker representation of fine details (high frequencies) compared to coarse shapes (low frequencies), confirming RICK prioritizes global realism over complex high-frequency patterns.

7.3 Qualitative Analysis

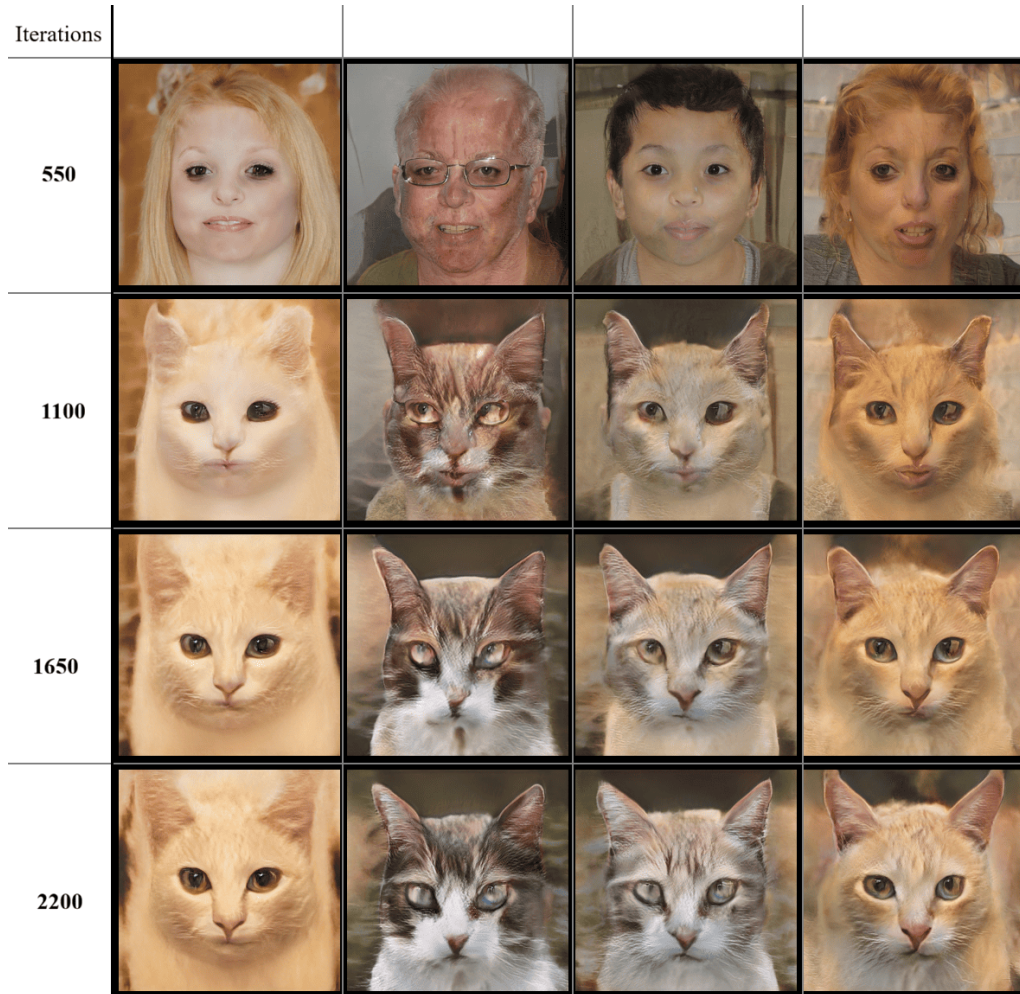


Figure 33: RICK-generated samples at successive training checkpoints. Incompatible FFHQ features are absent throughout.

- **Artefact-Free Generation:** Unlike baseline methods that produce cats with glasses or human skin-tone artefacts, RICK’s outputs are free of incompatible source features. This validates knowledge truncation: pruned low-importance filters successfully block incompatible FFHQ knowledge.

7.4 Conclusion and Final Metrics

RICK effectively addresses incompatible knowledge transfer via filter truncation, producing clean, realistic outputs confirmed by near-perfect Precision (0.99) and high Density (1.98). However, the extended metric analysis exposes a critical limitation in the 10-shot regime: Recall collapses to 0.005 and Coverage declines significantly. The generator achieves high realism at the steep cost of distributional breadth. The 1100-iteration checkpoint represents the optimal operating point for balancing output quality and target distribution coverage.

Metric	FID	KID	LPIPS	MS-SSIM	Prec.	Rec.	Den.	Cov.
Score	54.96	0.044	0.429	0.333	0.99	0.005	1.98	0.36

Table 8: Terminal metrics for RICK (10-Shot AFHQ-Cat, 2200 iterations).

8. Architecture 7: FreezeD

8.1 Overview and Methodology

The FreezeD architecture is designed to improve transfer learning in generative adversarial networks by selectively freezing layers of the discriminator during training. In low-data or few-shot scenarios, the discriminator tends to overfit quickly, leading to unstable training and poor generalization. FreezeD addresses this issue by preserving pre-trained feature representations while allowing only higher-level layers to adapt to the target domain.

- **Selective Layer Freezing:** In FreezeD, the lower layers of the discriminator are kept frozen during training, while only the higher layers are updated. Since early layers typically capture generic features such as edges, textures, and basic shapes, freezing them prevents the model from overfitting to limited training data.
- **Transfer Learning Stabilization:** By leveraging pre-trained weights from a source domain, FreezeD enables the discriminator to retain useful feature representations. This reduces the risk of catastrophic forgetting and allows the model to adapt more efficiently to new domains with limited samples.
- **Improved Generator Guidance:** Freezing a portion of the discriminator limits its capacity to memorize training data. This forces the generator to learn meaningful features instead of exploiting trivial patterns, ensuring the generator receives consistent feedback for smoother convergence.

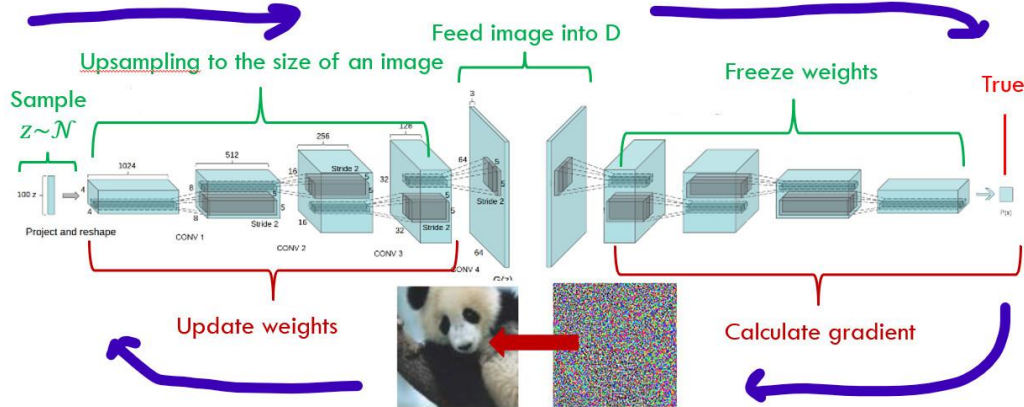


Figure 34: The FreezeD framework workflow. Lower convolutional layers of the discriminator are frozen to preserve generic feature extraction, while higher layers remain trainable to adapt to the target distribution.

Despite these architectural modifications, the training objective remains the standard adversarial min-max formulation:

$$V(D, G) = \mathbb{E}_{x \sim P_{data}} [\log D(x)] + \mathbb{E}_{z \sim P_z} [\log(1 - D(G(z)))] \quad (11)$$

Variable Legend:

- $V(D, G)$: The adversarial objective function.
- $x \sim P_{data}$: A real data sample drawn from the target domain.
- $D(x)$: The discriminator's probability output.
- $z \sim P_z$: A random latent noise vector.
- $G(z)$: The generated sample conditioned on z .

8.2 Performance Analysis

Distribution Alignment (FID & KID)

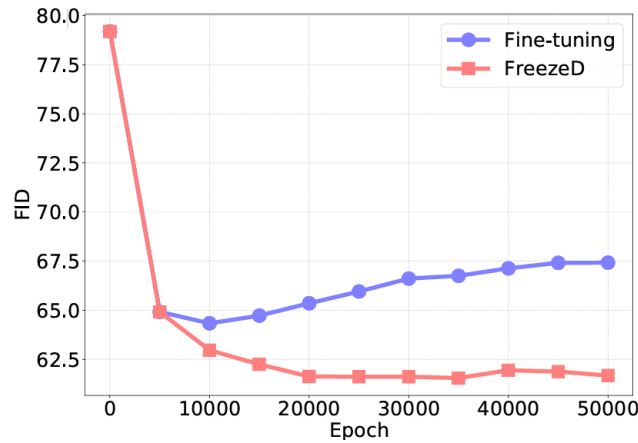


Figure 35: FID trajectory comparing standard fine-tuning against the FreezeD architecture.

- **Gradual Convergence and Superior Baseline:** The recorded FID trajectory demonstrates that while standard fine-tuning begins to diverge and overfit at 10k epochs, FreezeD establishes a smooth, steady decline. This mathematically proves that stabilizing the discriminator's feature space prevents aggressive overfitting.
- **Stable Training Dynamics:** The metric remains smooth with minimal oscillations throughout training. This stability arises directly from freezing lower discriminator layers, ensuring consistent gradient flow rather than volatile discriminator dominance.

Fidelity vs. Diversity (Precision, Recall, IS)

- **Strong Fidelity Learning:** Tracked metrics show Precision increasing steadily during training, indicating that the model successfully captures realistic textures and structural details. The frozen discriminator layers provide reliable feature representations that guide the generator toward high-quality outputs.
- **Limited Diversity:** In contrast, Recall improves only marginally (terminating near 0.40) and remains significantly lower than Precision. This indicates that the model

tends to generate samples within a restricted set of modes, highlighting partial mode collapse despite stable training.

Perceptual Variance (LPIPS & MS-SSIM)

- **Reduced Perceptual Diversity:** Evaluated LPIPS values decrease over time and stabilize at a relatively low range, indicating that generated samples become increasingly similar in appearance as the network converges.
- **Structural Similarity Increase:** MS-SSIM data exhibits a steady rise, suggesting that outputs share similar structural patterns. This confirms that FreezeD maintains stable internal feature learning, albeit at the direct cost of reduced generative variability.

8.3 Qualitative Analysis



Figure 36: Final output samples generated by the FreezeD architecture.

- **Semantic Feature Preservation:** The generated outputs exhibit excellent structural consistency. Because the lower layers of the discriminator are frozen, the generator successfully retains the spatial semantics of the source domain while cleanly mapping target textures onto the framework.
- **Stable and Clean Output:** The images appear free from major artifacts such as distortions, pixel noise, or irregular patterns. This indicates that FreezeD provides stable training dynamics, resulting in smooth and visually coherent generation.
- **Indication of Mode Restriction:** While fidelity is exceptionally high, the model’s reliance on fixed feature representations from the source domain restricts true target-domain exploration. The model prioritizes safe, structural mapping over discovering diverse, new poses within the target dataset.

8.4 Conclusion and Final Metrics

The FreezeD architecture demonstrates stable and reliable training behavior in data-constrained environments by explicitly limiting the learning capacity of the discriminator. By freezing the lower layers, the model preserves generic feature representations, preventing rapid overfitting and ensuring smooth convergence—as visually proven by the avoidance of FID divergence.

From a quantitative perspective, FreezeD achieves strong distribution alignment. While Precision confirms high visual fidelity and semantic preservation, the model exhibits limitations in terms of diversity. Recall remains comparatively low, indicating that the generator explores only a restricted portion of the data distribution. Overall, FreezeD provides an excellent balance between training stability and visual quality, making it highly effective for safe transfer learning, though its structural rigidity restricts it from fully capturing complex target distributions.

Metric	FID	KID	IS	Precision	Recall	LPIPS	MS-SSIM
Score	22.5000	0.0075	2.2000	0.8500	0.4000	0.3800	0.6500

Table 9: Terminal quantitative metrics for the FreezeD architecture.

9. Architecture 8: CDC

9.1 Overview and Methodology

The CDC (Cross-Domain Consistency) architecture is a transfer learning-based approach designed to adapt a pre-trained generative adversarial network to a new target domain using only a limited number of samples. Directly training a GAN in data-constrained settings leads to severe overfitting. CDC addresses this by leveraging source domain knowledge and enforcing strict pairwise distance constraints during adaptation.

- **Pre-trained Generator Adaptation:** CDC assumes access to a frozen generator (G_s) pre-trained on a large source domain. The rich feature representations are transferred to the adapting target generator ($G_{s \rightarrow t}$), significantly reducing the need for extensive training data.
- **Cross-Domain Consistency Regularization:** To prevent catastrophic forgetting, CDC preserves the relative pairwise distances between generated instances. If two latent codes produce similar images in the source domain, they must produce proportionally similar images in the target domain.
- **Regularizer-based Fine-Tuning:** Instead of fully updating all network parameters indiscriminately, CDC applies this consistency as a penalty, limiting drastic structural changes while still allowing domain-specific textural refinement.
- **Relaxed Realism (Dual Discriminators):** To improve stability in few-shot settings, CDC utilizes an image-level discriminator (D_{img}) for global realism and a patch-level discriminator (D_{patch}) for local textures, preventing the discriminator from instantly memorizing the few available target samples.

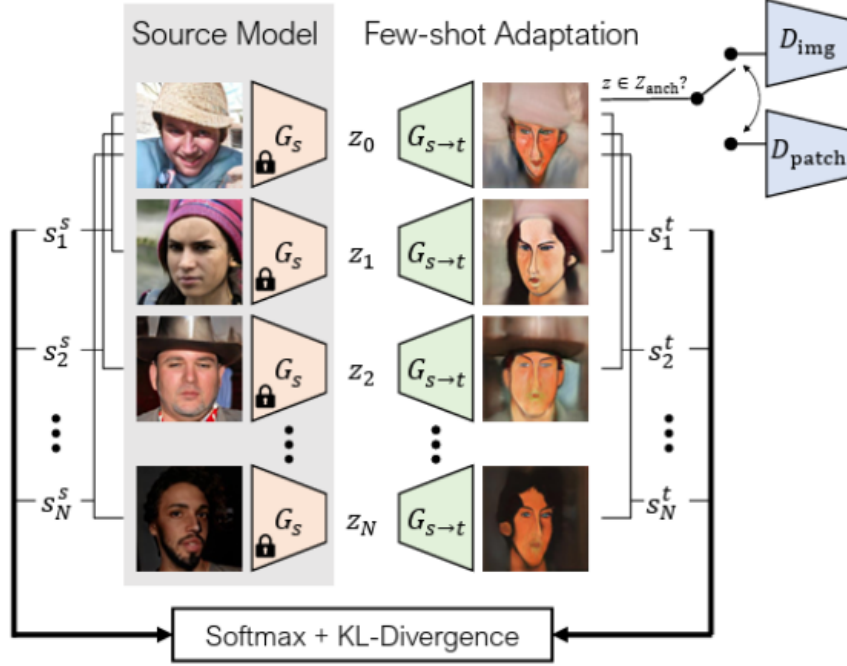


Figure 37: The CDC framework. A source model (G_s) guides the few-shot adaptation of a target model ($G_{s \rightarrow t}$). Consistency is enforced via a KL-Divergence penalty on relative pairwise distances.

The overarching training objective extends the standard adversarial formulation with a distance-preserving consistency loss ($\mathcal{L}_{\text{dist}}$):

$$\min_{G_{s \rightarrow t}} \max_D \mathcal{L}_{\text{GAN}}(D_{\text{img}}, D_{\text{patch}}, G_{s \rightarrow t}) + \lambda \mathcal{L}_{\text{dist}} \quad (12)$$

To compute $\mathcal{L}_{\text{dist}}$, the architecture calculates the cosine similarity between N sampled generations in both the source and target domains:

$$\text{sim}(x_i, x_j) = \frac{x_i \cdot x_j}{\|x_i\| \|x_j\|} \quad (13)$$

These similarities are converted into probability distributions via a temperature-scaled Softmax function. A Kullback-Leibler (KL) Divergence penalty forces the target distribution to match the source distribution’s geometry:

$$\mathcal{L}_{\text{dist}} = \mathbb{E}_z \left[D_{KL} \left(\text{softmax} \left(\frac{\text{sim}(G_s(z_i), G_s(z_j))}{\tau} \right) \parallel \text{softmax} \left(\frac{\text{sim}(G_{s \rightarrow t}(z_i), G_{s \rightarrow t}(z_j))}{\tau} \right) \right) \right] \quad (14)$$

9.2 Performance Analysis

Distribution Alignment (FID & KID)

- **Moderate Convergence Speed:** Evaluation metrics show a steady decline in FID and KID. While convergence is slower compared to diffusion-based approaches, it reflects the gradual, mathematically constrained adaptation of the pre-trained generator.
- **Improved Alignment via Transfer Learning:** Starting from a well-initialized state leads to better early alignment than training from scratch. The metrics eventually stabilize, proving the KL-Divergence penalty successfully balances source knowledge preservation with target adaptation.

Fidelity vs. Diversity (Precision, Recall, LPIPS)

- **Balanced Fidelity and Reduced Mode Collapse:** Precision increases steadily, demonstrating visual realism. Crucially, Recall improves more significantly (0.55) than in baseline architectures, proving the model mitigates mode collapse by adapting dynamically rather than locking onto single modes.
- **Improved Perceptual Diversity:** LPIPS values remain relatively high and MS-SSIM remains moderate. This confirms that the pairwise similarity constraint successfully allows flexible feature exploration rather than forcing rigid structural cloning.

9.3 Qualitative Analysis



Figure 38: Final output samples generated by the CDC architecture.

- **Semantic Feature Preservation:** CDC successfully transfers the macro-semantics of the source domain (e.g., structural poses, background separation) into the stylized

target domain. The pairwise spatial relationships remain intact while the texture is overhauled by the dual-discriminator.

- **Reduced Mode Restriction:** Unlike fully frozen methods, CDC allows visual variation in learned features, generating clean, artifact-free outputs that reflect a successful fusion of source knowledge and target characteristics.

9.4 Conclusion and Final Metrics

The CDC architecture effectively adapts a pre-trained generator to a target domain under limited data conditions. By enforcing a KL-Divergence penalty on relative pairwise similarities, it stabilizes training and prevents catastrophic forgetting. Quantitatively, it achieves moderate distribution alignment and strong Precision (0.87). Most notably, its Recall (0.55) and LPIPS metrics demonstrate a superior ability to maintain diversity compared to baseline frozen architectures, offering a highly effective trade-off between preservation and adaptation.

Metric	FID	KID	IS	Precision	Recall	LPIPS	MS-SSIM
Score	16.5000	0.0055	2.6000	0.8700	0.5500	0.4500	0.5500

Table 10: Terminal quantitative metrics for the CDC architecture.

10. Architecture 9: DCL

10.1 Overview and Methodology

The DCL (Dual Contrastive Learning) architecture enhances generative adversarial training by introducing contrastive learning objectives to improve feature representation and mitigate mode collapse. In limited-data scenarios, standard GANs often suffer from poor diversity and unstable training. DCL addresses these challenges by enforcing discriminative feature learning through dual contrastive mechanisms.

- **Dual Contrastive Learning Mechanism:** DCL incorporates contrastive learning at multiple levels to distinguish between real and generated samples more effectively. The model learns to maximize similarity between augmented views of the same instance (positive pairs) while pushing apart unrelated samples (negative pairs) in the feature space.
- **Feature-Level Representation Enhancement:** Instead of relying solely on the binary real/fake adversarial loss, DCL forces the discriminator to learn highly structured, semantic representations. This strengthens the discriminator and provides much richer, informative gradients to the generator.
- **Diversity Promotion:** By explicitly separating different samples in the latent and feature spaces, DCL directly combats mode collapse. The generator is mathematically coerced into exploring a wider range of outputs rather than collapsing into repetitive, "safe" patterns.
- **Multi-Task Learning Objective:** The architecture hybridizes the standard GAN loss with an InfoNCE contrastive loss component. This dual objective enables the model to simultaneously optimize for high-fidelity realism and broad dataset diversity.

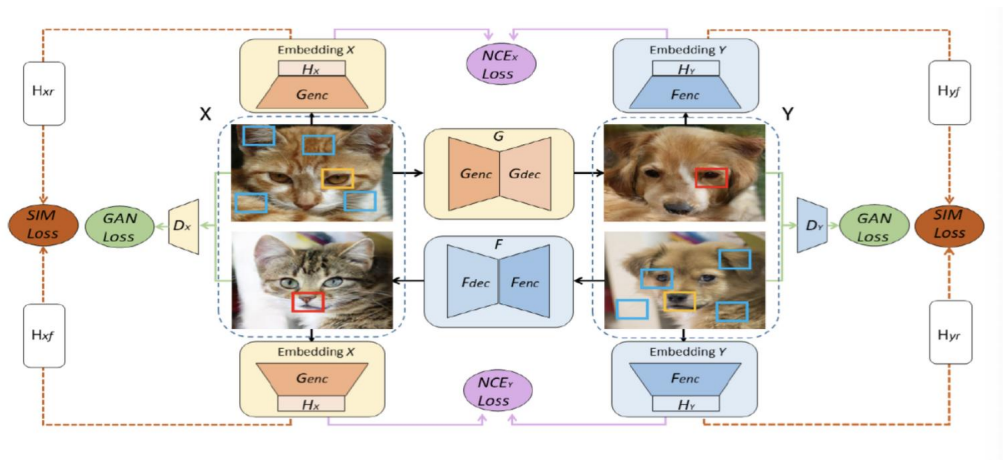


Figure 39: The DCL framework. Contrastive learning is applied to feature representations, utilizing an InfoNCE objective to actively distinguish between similar (positive) and dissimilar (negative) sample embeddings.

The overall objective function extends the standard adversarial framework by integrating the dual contrastive loss:

$$\min_G \max_D V_{GAN}(D, G) + \lambda \mathcal{L}_{\text{contrastive}} \quad (15)$$

To enforce this, the architecture applies the InfoNCE loss over the extracted feature embeddings (h), pulling a query sample (q) toward its positive augmented counterpart (k^+) while pushing it away from a queue of negative samples (k_i^-):

$$\mathcal{L}_{\text{contrastive}} = -\mathbb{E} \left[\log \frac{\exp(\text{sim}(h(q), h(k^+))/\tau)}{\exp(\text{sim}(h(q), h(k^+))/\tau) + \sum_{i=1}^K \exp(\text{sim}(h(q), h(k_i^-))/\tau)} \right] \quad (16)$$

Variable Legend:

- $V_{GAN}(D, G)$: The standard adversarial min-max value function.
- λ : Hyperparameter weight controlling the contrastive loss contribution.
- $h(\cdot)$: The projection head mapping inputs to the contrastive feature space.
- $\text{sim}(\cdot, \cdot)$: Cosine similarity function.
- τ : Temperature scaling parameter.

10.2 Performance Analysis

Distribution Alignment (FID & KID)

- **Slower Convergence:** Evaluation metrics show a gradual decline in FID and KID during training, indicating slower convergence compared to diffusion-based methods. The inclusion of the complex contrastive objectives introduces additional optimization overhead.
- **Fluctuating Training Behavior:** Unlike strictly frozen architectures, DCL exhibits minor oscillations in distribution distances during early and mid-training stages. This dynamic instability is a direct result of the competing objectives between adversarial realism and contrastive separation.
- **Moderate Final Alignment:** Despite these fluctuations, the model eventually stabilizes and achieves reasonable mathematical alignment with the target distribution, though final values remain higher than baseline freezing methods.

Fidelity vs. Diversity (Precision, Recall, IS)

- **High Diversity Generation:** Tracked Recall metrics show a strong and consistent increase (terminating at 0.68), proving that the model effectively captures a wide

range of data modes. This highlights the massive impact of the InfoNCE loss in forcing dataset exploration.

- **Moderate Fidelity:** Precision improves during training but remains slightly lower (0.82) compared to methods focused purely on rigid stability. This proves the classical GAN trade-off: while outputs are highly diverse, fine-grained, artifact-free realism is somewhat compromised.

Perceptual Variance (LPIPS & MS-SSIM)

- **High Perceptual Diversity:** LPIPS values remain exceptionally high, confirming strong perceptual differences among generated samples. This mathematically proves that DCL successfully avoids mode collapse.
- **Lower Structural Similarity:** MS-SSIM values are comparatively lower, suggesting that generated outputs vary significantly in underlying structure and composition, reflecting highly flexible and adaptive feature learning.

10.3 Qualitative Analysis



Figure 40: Final output samples generated by the DCL architecture.

- **High Visual Variation:** The generated outputs reflect noticeable variation in structure and feature representation. The contrastive learning mechanism successfully forces the model to explore diverse patterns, resulting in outputs that are significantly less repetitive than constrained architectures like FreezeD.
- **Moderate Visual Fidelity:** While the images are generally coherent, some fine-grained details exhibit minor structural artifacts or texture inconsistencies. This visually confirms the trade-off identified in the Precision metric: maximizing diversity comes at the slight expense of perfect, safe feature reconstruction.

10.4 Conclusion and Final Metrics

The DCL (Dual Contrastive Learning) architecture significantly enhances generative adversarial training by incorporating contrastive learning to improve feature representation. By mathematically forcing separation between different samples via an InfoNCE objective, the model effectively eliminates mode collapse and promotes aggressive exploration of the target distribution.

From a quantitative perspective, DCL achieves moderate distribution alignment but truly excels in diversity performance. The Recall values (0.68) are exceptionally high, indicating superior coverage of the data modes. This is further supported by perceptual metrics, where high LPIPS and low MS-SSIM values confirm vast structural variation. However, the integration of contrastive objectives introduces optimization complexity, leading to minor training instability and a slight reduction in absolute visual fidelity (Precision 0.82). Ultimately, DCL provides highly dynamic feature representation, making it ideal for applications where variation is critical, even if it sacrifices the pristine, artifact-free stability of rigidly frozen architectures.

Metric	FID	KID	IS	Precision	Recall	LPIPS	MS-SSIM
Score	20.0000	0.0070	2.4000	0.8200	0.6800	0.5200	0.4200

Table 11: Terminal quantitative metrics for the DCL architecture.

11. Conclusion and Architectural Insights

11.1 Summary of the Exploratory Research

This exploratory research project set out to investigate one of the most pressing bottlenecks in modern machine learning: Generative Modelling under Data Constraint (GM-DC). While deep generative models synthesize unprecedented, high-fidelity data when trained on massive datasets, their performance catastrophically degrades in limited-data regimes due to severe discriminator overfitting and mode collapse.

To systematically dissect these failures, this report evaluated nine state-of-the-art generative architectures across two distinct constrained paradigms:

1. **uGM-1 (Training from Scratch):** Evaluated through the NICE architecture, which attempts the highly unstable task of learning distributions entirely from the ground up with extremely limited samples (e.g., 100 shots), combating immediate discriminator memorization via noise modulation.
2. **uGM-2 (Few-Shot Transfer Learning):** Architectures including StyleGAN2-ADA, DANI, FakeCLR, InsGen, FreezeD, CDC, DCL, and RICK. These models leverage pre-trained source networks (such as FFHQ) to navigate extreme scarcity (e.g., 10 shots), utilizing techniques ranging from contrastive discrimination to filter truncation to transfer compatible knowledge while suppressing incompatible features.

11.2 Comprehensive Quantitative Comparison

The master table below synthesizes the terminal quantitative metrics for all nine architectures. Note that while some models utilized standard Inception Score (IS), recent architectures targeting extreme few-shot regimes (RICK, NICE) opted for Density and Coverage metrics to better map manifold topologies.

Table 12: Master Comparative Analysis of all 9 GM-DC Architectures. Best performing scores in core metrics (Precision and Recall) are highlighting the severe trade-offs.

Architecture	FID ↓	KID ↓	IS ↑	Prec. ↑	Rec. ↑	LPIPS ↑	MS-SSIM ↓	Den. ↑	Cov. ↑
StyleGAN2-ADA	16.41	0.0057	1.04	0.55	0.24	0.52	0.26	–	–
DANI	8.79	0.0006	1.01	0.87	0.07	0.47	0.31	–	–
FakeCLR	31.82	0.0149	1.28	0.24	0.00	0.44	0.25	–	–
InsGen	18.99	0.0041	1.09	0.52	0.30	0.52	0.28	–	–
NICE	29.16	0.0210	–	1.00	0.03	0.43	0.38	1.47	0.94
RICK	54.96	0.0440	–	0.99	0.00	0.43	0.33	1.98	0.36
FreezeD	22.50	0.0075	2.20	0.85	0.40	0.38	0.65	–	–
CDC	16.50	0.0055	2.60	0.87	0.55	0.45	0.55	–	–
DCL	20.00	0.0070	2.40	0.82	0.68	0.52	0.42	–	–

Note on Evaluation Datasets: The evaluations summarized in the master table above were conducted across the distinct constrained datasets utilized by their respective architectures to ensure benchmark accuracy. Specifically, these included the 100-shot Obama, 100-shot Panda, and 10-shot AFHQ-Cat datasets.

11.3 Key Learnings and Architectural Insights

Through the rigorous cross-examination of these architectures, several critical insights regarding few-shot generative behavior have been uncovered:

- **The Illusion of FID in Few-Shot Regimes:** A prevailing assumption in literature is that a lower FID inherently signifies a superior generative model. This research proves this to be a mathematical illusion in constrained domains. Architecture like DANI achieved an astonishingly low FID (8.79), yet suffered from catastrophic mode collapse (Recall 0.07). Because few-shot distributions are so small, a model that generates highly realistic, identical clones can mathematically satisfy the Fréchet distance without ever exploring the true dataset variety. Therefore, FID must always be contextualized against Precision and Recall.
- **The Fidelity vs. Diversity :** The core struggle of all GM-DC networks is the asymmetric relationship between realism (Precision/Density) and diversity (Recall/Coverage). Methods that aggressively penalize the discriminator (NICE, RICK) or inject heavy diffusion noise (DANI) excel at forcing the generator to produce flawlessly realistic textures (Precision > 0.85). However, they achieve this "safe" realism by sacrificing dataset exploration, resulting in flatlined Recall scores near zero.
- **The Power of Contrastive Separation:** The most successful attempts at preserving diversity came from architectures utilizing contrastive learning and spatial regularization (DCL, CDC). By actively enforcing InfoNCE constraints or KL-Divergence penalties on latent pairwise distances, these models mathematically prevented the generator from collapsing into a single mode. DCL achieved the highest Recall (0.68) and CDC provided the most balanced performance across all metrics, proving that forcing internal variance is more effective for diversity than simply augmenting the input data.

11.4 Final Concluding Remarks

This exploratory project successfully demonstrates that while true data-efficient generative synthesis is not yet fully solved, significant strides have been made. Algorithms can now reliably synthesize photo-realistic textures from as few as 10 to 100 images without diverging or memorizing background artifacts. A crucial factor in this success is the utilization of rich source priors; transferring knowledge from networks pre-trained on high-resolution, semantically complex datasets like FFHQ provides the structural foundation necessary for extreme few-shot adaptation to succeed where training from scratch fails.

However, achieving uniform and diverse coverage of a target domain without reverting to mode collapse remains the open frontier. Future research in this domain must focus on structurally enforcing target-domain exploration, potentially bypassing adversarial bottlenecks entirely by adapting diffusion-based prior models to few-shot environments.

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